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Fisher Controls

DPR900 Integral Controller User Manual

This Manual Supports Software Version 8836 thru 9147

D4U00791022

User Manual
UM6.1:DPR900:9147
May 1991
Change 1 – September 1991
Change 2 – July 1992

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DPR900 Integral Controller User Manual

This Manual Supports Software Version 8836 thru 9125

User Manual
UM6.1:DPR900:9102
May 1991
Change 1 – September 1991

D4U00792012

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How to Insert This Change

1. Remove title page and insert Change 1 title page.
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5. Remove page 2–9/2–10 and insert changed page 2–9/2–10. Reason for change was to clarify connection of analog output signals.
6. Remove page 3–13/3–14 and insert changed page 3–13/3–14. Reason for change was to remove unnecessary text.
7. Remove page 5–1/5–2 and insert changed page 5–1/5–2. Reason for change was to add information in table.
8. Remove page 5–7 thru 5–10 and insert changed page 5–7 thru 5–10. Reason for change was to add clarifying text and change table.
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10. Remove page 5–25/5–26 and insert changed page 5–25/5–26. Reason for change was to add information in table.
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12. Remove page 5–37/5–40 and insert changed page 5–37/5–40. Reason for change was to add information to table and expand text.
13. Remove page 5–57 thru 5–64 and insert changed page 5–57 thru 5–64. Reason for change was to add table on 5–58 and correct table information as highlighted.
14. Remove Appendix B and insert changed Appendix B. Reason for change was to add FCODE's to index.
15. Remove Appendix C and insert changed Appendix C. Reason for change was to correct part number and add new part.
16. Remove end page and insert new end page. Reason for change was to correct informational block.
17. Place this instruction page behind Change 1 Page A, as a record of Change 1 insertion.

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Change 2 Instructions for ...

UM6.1:DPR900:9102
DPR900 Integral Controller
User Manual
(Original — May 1991)
(Change 1 — September 1991)

The following information and attachments constitute Change 2 to the DPR900 Integral Controller User Manual, UM6.1:DPR900:9102. Vertical change bars are used to highlight the material affected by this change.

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4. Remove page 1–3/1–4 and insert changed page 1–3/1–4.

   Page 1–4 has an addition to the digital output signals for ac power.
5. Remove page 2–3/2–4 and insert changed page 2–3/2–4.

   On page 2–4 the power connections for an ac-powered DPR910 are relocated.
6. Remove pages 2–11 thru 2–16 and insert changed pages 2–11 thru 2–16.

   A note regarding power usage has been added to page 2–11. The 38V value is added to the drawing on page 2–12. Subsection 2.13 on page 2–14 is upgraded.
7. Remove pages 5–17 thru 5–22 and insert changed pages 5–17 thru 5–22.

   On page 5–17, a notation has been added to the end of subsection 5.6.3.1 regarding D03 and D04. On page 5–19, a sentence has been added at the end of the page. On page 5–20, a footnote has been added to the FCODE 103 options and the next to last paragraph has been changed. A sentence has been added near the top of page 5–21.

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8. Remove page 5–33/5–34 and insert changed page 5–33/5–34.

   The drawing on page 5–34 has had the D03 deactivated text added.

9. Remove page 5–39/5–40 and insert changed page 5–39/5–40.

   On page 5–39, a note is added regarding reversing the output signal display.

10. Remove pages 5–57 thru 5–64 and insert changed pages 5–57 thru 5–64.

    Page 5–57 has a change to the version number in subsection 5.17. Page 5–58 also has a version number change. Page 5–60 in Worksheet 5–2, the *Calculation of Process Variable* function has been changed to show AI1 as the first variable in the last two formulas. On page 5–63, changes are made to FCODE 135 and 136.

11. Remove Appendix A page A–1/A–4 and insert changed Appendix A page A–1/A–4.

    Fault code E110 is added on page A–3.

12. Place this instruction page behind Page B, Change 2 List of Effective Pages, as a record of Change 2 insertion.

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1 Introduction

1.1 Scope

This manual provides information required for installation, operation, and configuration of the DPR900 Integral Controller.

1.2 Description of Manual

Section 1 = Introduces the user to the organization of the manual, contents of other sections, and DPR900 Controller specifications.

Section 2 = Describes the mechanical and electrical installation procedures for the controller, including power and grounding requirements.

Section 3 = Explains the functions and use of the front panel controls and indicators.

Section 4 = A brief description of the procedures used to set FCODE's.

Section 5 = Instructions for configuring the DPR900 Controller.

Appendix A = Fault Codes

Appendix B = FCODE Index

Appendix C = Field Replaceable Parts

1.3 Controller Description

The DPR900 Integral Controller is a single-loop, proportional, integral, derivative (PID), user-configurable controller. All controller functions reside in the controller. Configuration consists of selecting the appropriate functions and setting the tuning parameters, using the controls and indicators on the front panel. Configuration may be entered or changed while the controller is operating. The controller scans and updates all input and output signals five times a second. When the controller is communicating with a computer via the RS-485 serial interface, this I/O scan and update rate is not valid, due to communications link limitations. Connection to and communication with external equipment is via four analog and four digital inputs, two analog and six digital outputs, and a serial data port (RS-485).

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The controller may communicate with a central computer system via the RS-485 serial data communication channel. The communications mode is half-duplex, using the COMLI communications protocol.

Two discrete inputs may be used to directly control either the output signal or setpoint value. Other discrete inputs can switch the controller to a remote or extra internal setpoint or a predetermined value. These external discrete signals may also switch the controller between different modes: Auto/Manual or Local/Remote.

The terminal block, which connects all input power and signal lines, is at the back of a separate, permanently installed mounting cassette. The controller slides into this cassette, eliminating the need to disconnect cables or disturb the electrical installation when changing a controller.

A configuration reference card is located in the slot behind the hinged flap at the top of the controller's front panel. It lists the controller functions and tuning parameters. Space is provided for the user to indicate the configuration and tuning values currently entered in the controller.

DPR900 Specifications

CONTROL FUNCTIONS

PID function as standard

Manual or autotuner PID parameter setting

Gain scheduling via as many as three different parameter settings

Feedforward function selection

CONTROL RANGE

Amplification (Gain) between 0.01 and 99.99

Integral time (Reset) between 0.1 and 9999 seconds or disabled (OFF)

Derivative time (Rate) between 0 and 9999 seconds

Direct or reversed control action

Full alarm and fault indications.

CONTROLLER FUNCTIONS

Process Value

Linearization, filtering, and direct temperature inputs for a Pt-100 RTD or Type J, K, R or S thermocouple

User-configurable input signal range

Process variable calculation function

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Process variable monitoring function

Selectable analog and/or digital display

Min/Max process variable signal selector

Setpoint

Internal, extra internal, external, or computer set

Setpoint limits digitally set

Setpoint ramp and hold function

Remote setpoint with ratio and bias functions

Setpoint tracking function

Setpoint calculation function

Analog and/or digital indication

Min/Max setpoint signal selector

Output Signal

Selected output types:

- Analog
- Pulse-width modulated
- Positioning motor control

Output signal range limited, computer-controlled, or controlled with digital input signals

Selectable start-up and default values

Can be forced to hold at a pre-set value

Reversible analog indication

Output signal detection (malfunction)

Min/Max output signal selector

Alarms

Process value or deviation alarms

Loss of input signal alarm

Process value monitoring alarms

Loss-of-output signal alarms

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Automatic self-testing alarms

Incorrect configuration alarms

Miscellaneous

External output of selected input signal

Input signal filtering and linearization

External Auto/Manual and Remote/Local switching

Selectable digital signal logic

Digital output indication of mode, alarms, and panel signals

On-line entry and alteration of configurable functions

INPUT SIGNALS

Analog

Four floating analog inputs, allowing signals between –5 volts and +19 volts relative to supply ground. +5 volts signal added to input voltage giving 19 + 5 = 24 volts, which can vary by ± 10%

Signal range: 4–20 mA (1–5V) or 0–20 mA (0–5V)

Digital

Four optically-isolated digital inputs for 24 Vdc.
Logical 0: interpreted as 0-5 volts
Logical 1: interpreted as 16-30 volts

Input Resistance

Current inputs: 250 ohms
Voltage inputs: greater than 500K ohms

OUTPUT SIGNALS

Analog

Two analog 4–20 mA or 0–20 mA

Load 0 to 650 ohms

Load dependency less than 0.2% with a change from 0 to 650 ohms

Digital

Four digital outputs of 24V, up to 250 mA each (DO3—DO6, with external 24V power only).
AC-powered, maximum of 100 mA total for all outputs.

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Two optically-isolated digital outputs for 24V/20 mA (DO1, DO2)

Output Signal Failure

Output signal line failure alarm

Power Failure Restart

Restart from preselected value or last values

PROCESS INDICATORS

Process/Setpoint Values

Analog indication on vertical bargraph displays with 2.5% resolution (0-100% scale)

Digital indication on five-character display in engineering units. Process variable is default display

Output Signal

Analog indication on vertical bargraph display with 5% accuracy, graduated 0-100%. During manual control, resolution is 0.5%

Can be used with positioning motor output control to indicate actuator positioning

Alarm and Fault Indicators

Shown on five-character display and two digital outputs.

Miscellaneous

Computer control indicated by **C** in display

Setpoint indicated by **SP** shown on display

During configuration, function codes and tuning parameter values appear on display

Functions chosen are indicated by LED's illuminated on pushbuttons

ACCURACY

Maximum error less than 0.2% of measuring range.

Thermocouple

Less than 0.01% of thermocouple measuring range per °C within the range of 0-50°C ambient. Accuracy is 10-20°F (5-10°C) for Type R and S thermocouples.

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RTD

3°F (1.5°C) for 100 ohm Platinum RTD

Scan Rate

All inputs and outputs scanned and updated five times per second (200 ms)

POWER REQUIREMENTS

AC Power

110/120 or 220/240 Vac, ± 10% at 50/60 Hz. Power consumption 10VA,17VA maximum.

DC Power

24 Vdc ± 10%, 7.2W maximum, excluding I/O circuits

External Load

24 Vdc ± 10%, 100 mA total with ac supply. Three-stage pulse output supplies up to 20 mA to potentiometer (750 ohms).

ENVIRONMENTAL REQUIREMENTS

Temperature

32 to 122°F (0 to 50°C) in operation.

Installation

Flush fitting in mounting cassette, installed with retaining springs from the front panel.

Environmental Classification

In accordance with IEC 529, IP20 in all directions (meets or exceeds NEMA 12), IP65 from the front when mounted in a sealed panel

Electrical Connections

Screw terminals will accommodate up to 14 AWG ($1.5mm^2$) wire. All connections located in terminal blocks on the rear of the mounting cassette.

Computer Communications

Standard RS-485 serial interface (COMLI protocol)

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Electrical Safety

SEN 436 15 03
CSA certified (non-hazardous locations)

Static Discharge

SEN 436 15 22 (15k volts).

DIMENSIONS

Width: 2.84 inches (72 mm)
Height: 5.67 inches (144 mm)
Length: 10.83 inches (275 mm)

WEIGHT

5.5 pounds (2.5 kg)

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2 Installation

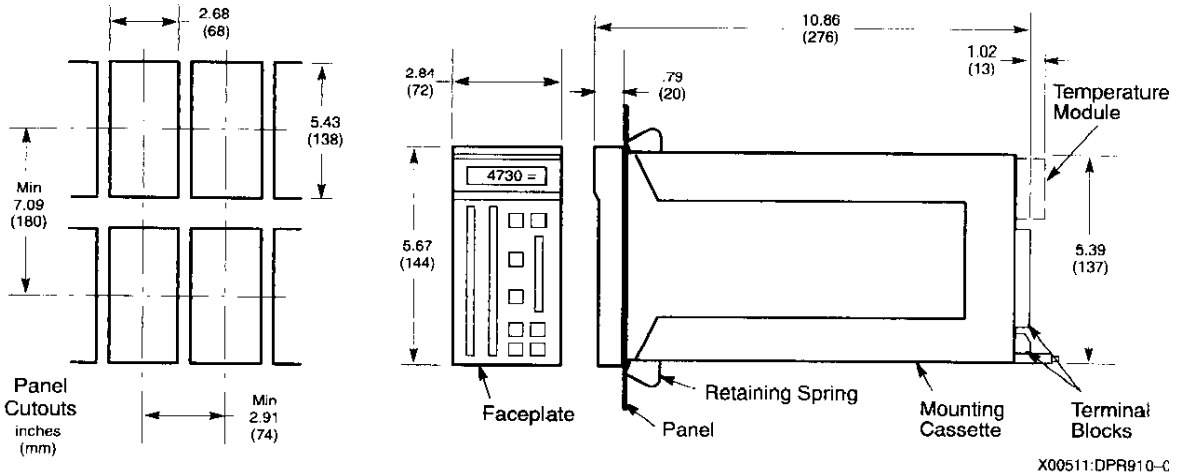
2.1 Mechanical Installation

The DPR900 Integral Controller units fit into a cutout in a vertical panel, allowing for high packing density. The cutout may be for a single controller unit or for multiple units mounted side by side. For typical installations the panel must be between 0.08 inch (2 mm) and 0.16 inch (4 mm) thick. For installation information on thicker panels, contact your Fisher Representative.

The figure shows a controller's external dimensions, along with the spacing and dimensions for panel cutouts. Front panel cutouts conform to DIN standards.

Caution

The panel must be able to support the weight of all controllers mounted in it. A large number of controllers in a panel may require additional panel support.



DPR900 Mounting Cassette Dimensions

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Note

Controllers cannot be installed vertically in the same cutout. Cutouts should not be closer than shown above.

Make the cutouts in the panel according to the dimensions and spacing shown. For multiple installations, make the cutout wider according to the following formula:

Cutout width (W) = N x B – 0.19 inch (4.5 mm)

where:

N = number of controllers to be mounted
B = unit width = 2.84 inches (72 mm)
Tolerance = ± 0.04 inch (1 mm) per unit

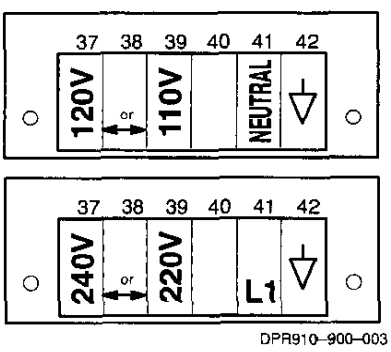
2.2 Wiring the Mounting Cassette

The DPR900 Integral Controller fits into a standard panel-mount cassette. The mounting cassette has terminal blocks on the rear for all electrical connections.

Connect all wiring to the terminal blocks before inserting the cassette into the panel cutout unless the rear of the cassette will be accessible after installation.

2.3 Power and Ground Wiring

The power source for the DPR900 Controller should be free of electrical noise. Do not use sources that supply power to relays, contacts, or thyristor-controlled equipment. If no other source is available, use an isolation transformer or magnetic voltage stabilizer. Do not connect electrically noisy equipment and the DPR900 to the same secondary of such a transformer. Closely follow the manufacturer's recommendations concerning the grounding of the isolation transformer.



DPR910-900-003

When operating the controller from an alternating current source, always use a 3-wire power cord. Connect the neutral wire to terminal 41 and the ground wire to terminal 42 on the terminal block. Connect the supply line wire to the appropriate terminal for the power being input (110, 120, 220, 240V). The controller operates on either 50 or 60 hertz power.

When the controller is operating from an ac power source, the 24 Vdc terminals (35 and 36) may be used to supply dc power to field instruments.

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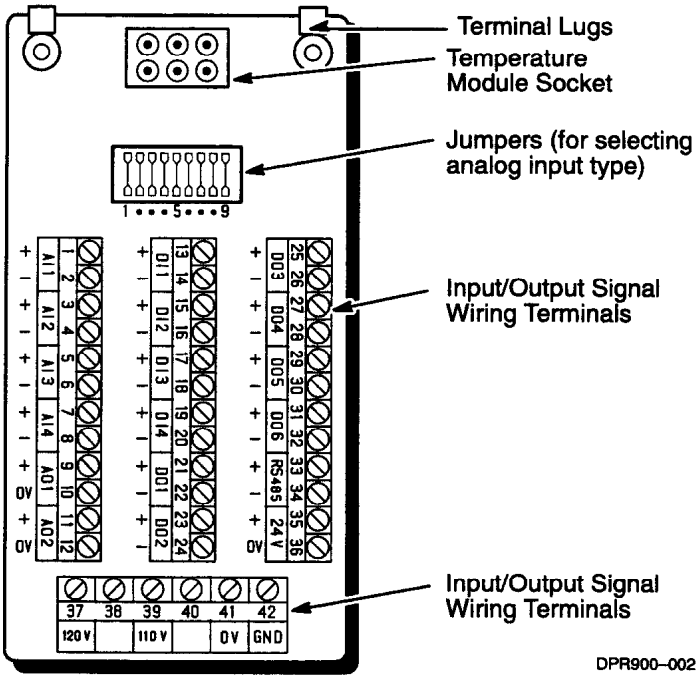
For proper operation of DPR products, it is essential that ground wiring be installed as indicated. To reduce electrical noise and interference and increase safety, connect separate ground wires from each unit to the Local Ground Bus (LGB) in the cabinet or enclosure. The ground wire from the LGB to the plant ground should be run in plastic or other non-ferrous material conduit. Ground connection of the same type, i.e. dc common connections, may be to the same point on the LGB. Never intermix dissimilar grounds.

Terminal 36, the dc common and Terminal 42, the case ground, should always be connected to the LGB. The remaining connections are dependent on type of input power (AC or DC).

If 14 or 16 AWG wire is used to supply AC power, the incoming ground wire should terminate at the LGB. A second wire should be run from the termination point on the LGB to Terminal 42. If a smaller wire size is used, connect the incoming ground wire to Terminal 42. Then run a second wire from Terminal 42 to the LGB.

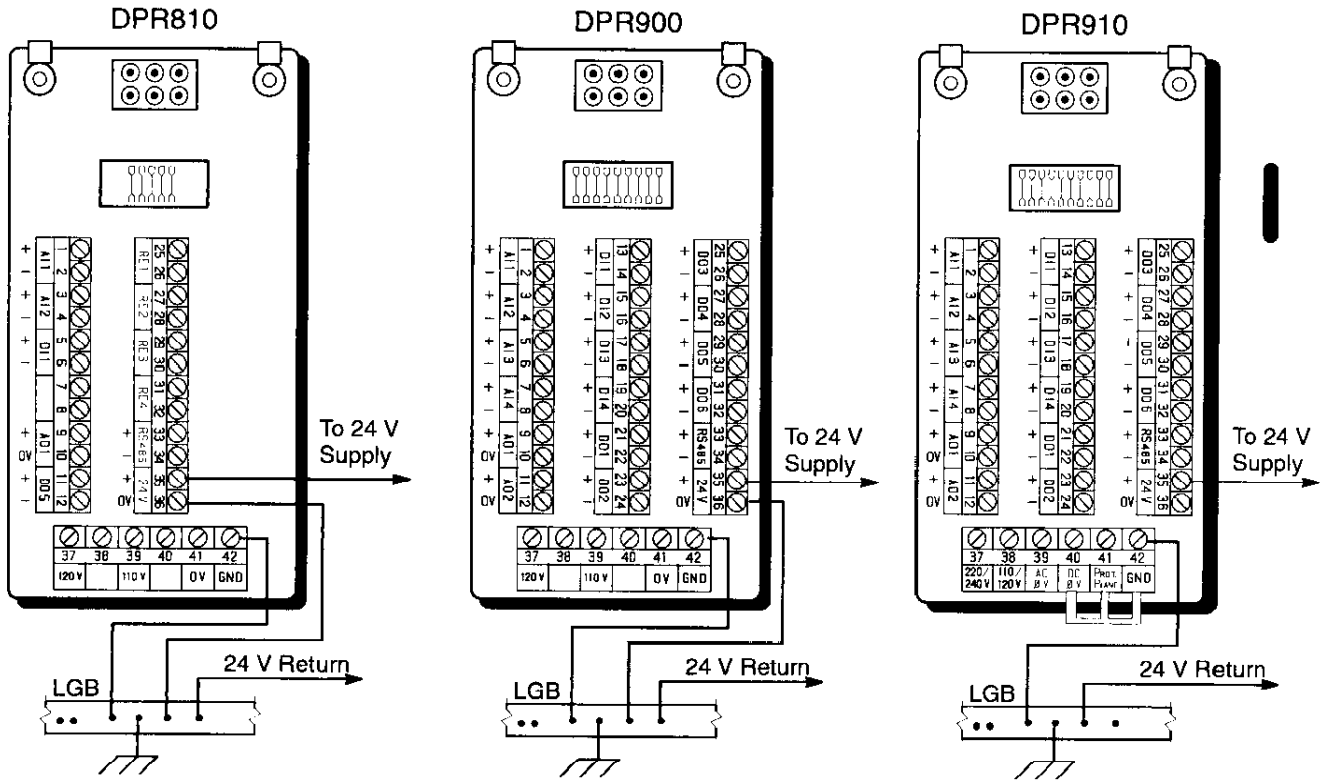
When considering the incoming wire size, ensure that any line loss does not exceed the voltage tolerance of the DPR unit. A power conditioner may be required.

If the unit is mounted in a metal panel or enclosure, the panel or enclosure must be grounded to the DPR. Use the terminals lugs at the rear of the mounting cassette for this purpose.

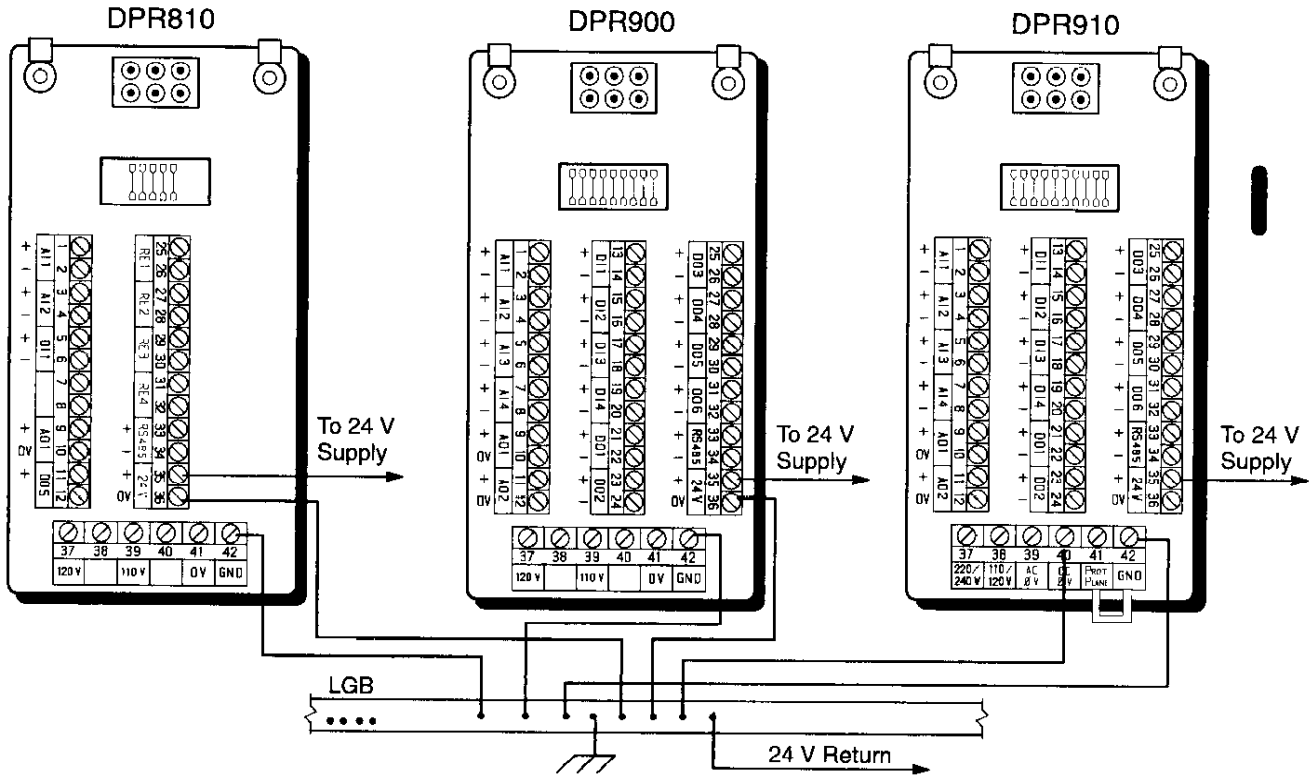


Terminal Block Connections Diagram

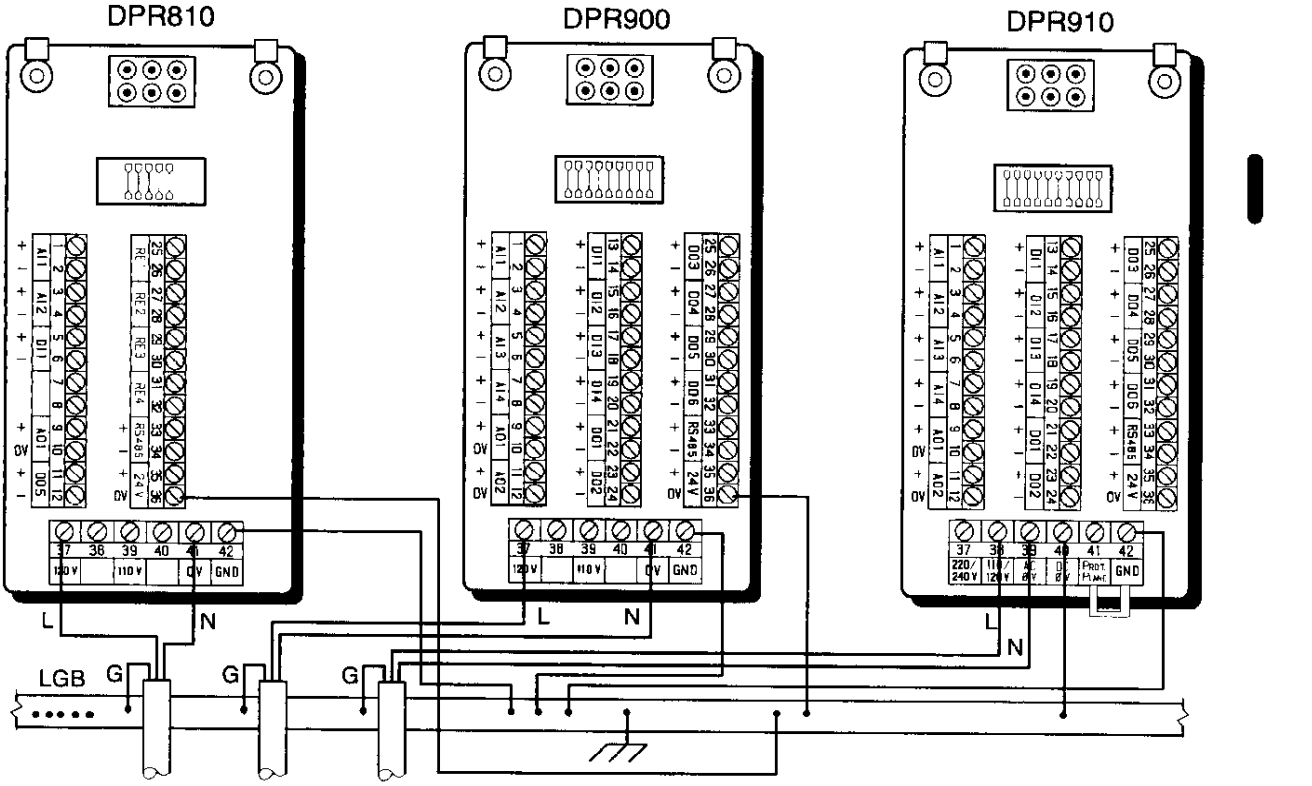
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Multi Unit, DC Powered



Multi Unit, AC Powered

DPR900-057

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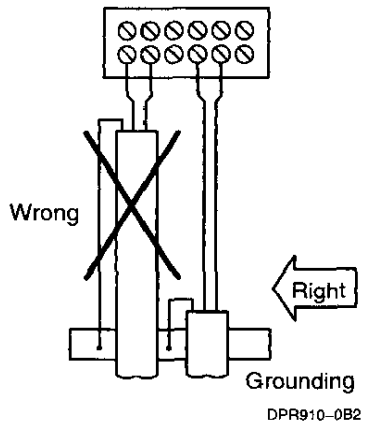
Note

The 24 Vdc terminal can supply a maximum external load of 100 mA when powered from 110, 120, 220 or 240 Vac power sources. This supply is intended to power transmitters or relays connected to the discrete outputs. The 100 mA limit applies only to ac powered units. No internal load, including the two analog outputs, need be included in the 100 mA total.

Controllers powered from 24 Vdc sources have a current limit of 250 mA on each output.

To operate the controller from a 24 Vdc supply, connect the power supply leads to 24 V terminals 35 and 36 (observe the proper polarity).

2.4 Signal Wiring



Connect all input and output wiring to the terminal blocks at the rear of the mounting cassette. Use direct runs of individually shielded, twisted-pair wire in an insulating jacket. The maximum wire size for terminal connectors is 14 AWG (1.5mm^2). The insulating jacket should only be stripped to allow the length of wire required to connect the twisted-pair to the terminals.

For long runs, it may be practical to route field wiring to a junction box, then through multi-pair shielded cable. Lace wires as required by local electrical codes or plant standards. Connect the shield and drain wires to the grounding bus bar as shown. Standard practice is for shields to be grounded at the signal source only. If the shield is grounded at the DPR (vs the external instrument), also tie it to the LGB. Shield wires should not be stripped further than one inch from the end of the signal wires.

Route a 14 AWG (1.5 mm^2) conductor from the mounting cassette ground to the panel ground point. Then use an AWG 8 (8.37 mm^2) multistrand conductor or 0.50 inch (12.7 mm^2) braided ground conductor (preferred) to connect the panel ground point to the LGB. The panel ground point and the ground of the ac power source for the unit should be connected to the same grounding bus bar. This minimizes the possibility of significant voltage between ground points. Use crimped or thermal-welded connections at all ground points.

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Route a 14 AWG (1.5 mm^2) conductor from terminal 36 to the instrumentation system dc ground point. Then run a 10 AWG (5.26 mm^2) multistrand conductor from the instrumentation system dc ground point to the existing plant ground grid. This point on the plant ground grid must be dedicated exclusively to the instrumentation system dc ground and must not be used for other equipment grounding. Use crimped or thermal-welded connections at all ground points.

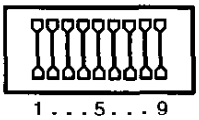
2.5 Jumper Module

Caution

Ensure that power to the controller is OFF, prior to cutting jumpers. Leave sufficient metal to solder jumper back together if required.

The jumpers on the back of the mounting cassette determine the electrical termination characteristics of the four analog inputs and the RS-485 communications wiring. All of the analog inputs, AI1 thru AI4, have 250 ohm input resistors. There is also a 100 ohm termination resistor on the RS-485 connection. The figure shows the jumper module and the table describes the connection each jumper makes.

Jumper #	Assignment	Closed	Open
1	Input resistor AI1	Current In	Voltage In
2	Input resistor AI2	Current In	Voltage In
3	Input resistor Ai3	Current In	Voltage In
4	Input resistor AI4	Current In	Voltage In
5	Floating ground AI1	Fixed GND	Floating GND
6	Floating ground AI2	Fixed GND	Floating GND
7	Floating ground AI3	Fixed GND	Floating GND
8	Floating ground AI4	Fixed GND	Floating GND
9	Termination for RS485 link	YES	NO



DPR900-58a

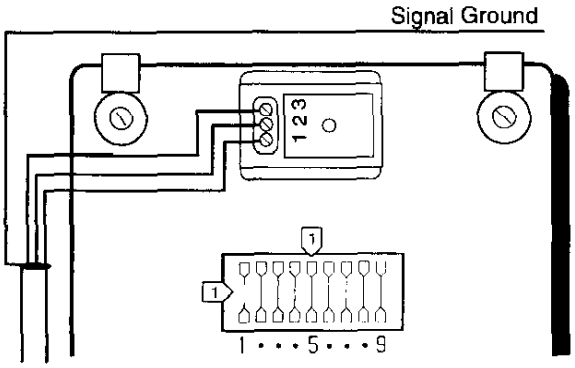
Jumpers 1 thru 4 connect the 250 ohm input resistors in series so that the 4-20 mA input signal can be converted to a 1-5V signal internally. Cut the appropriate jumper to accept a direct 1-5V signal on the input terminals. Jumpers 5 thru 8 connect inputs AI1 thru AI4 to the controller supply ground. Cut the appropriate jumper to allow the negative input terminal to float between –5 V and 17 V, with respect to supply ground.

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Jumper 9 is used to terminate the last controller on the RS-485 communication link. When the link is used, this jumper must be cut on all but the last controller on the link. Otherwise, it should remain intact.

2.6 Low-Level Temperature Input

A DPR100 Series Temperature Input Module may be plugged into the controller to allow direct connection of a low-level input signal. The controller internally connects this input to AI1. This disables AI1 for other uses. For further information on the DPR100, refer to Fisher Controls Bulletin 6.1:DPR100. The figure illustrates the wiring of the temperature input module.



Notes:

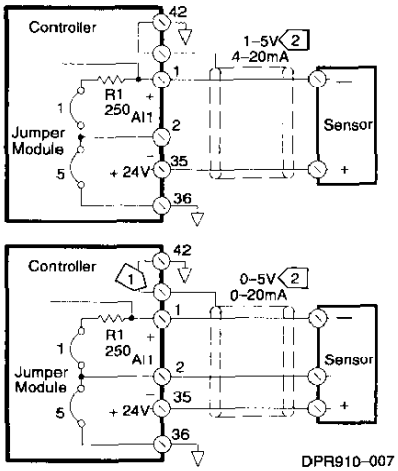
(1) When a temperature input module is connected to AI1, jumper 1 must be cut, and Jumper 5 must be intact. A floating ground (zero) is not allowed. DPR900-006

Note

FCODE 119 **must** be set to Option 2 (0-20 mA, 0-5 V) if a DPR100 Series Temperature Input Module is used.

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2.7 Analog Input Signals

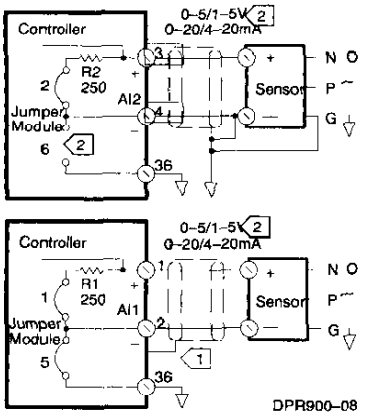


Notes:

1. Chassis ground
2. For voltage inputs the resistance jumper must be cut.

Connect analog inputs AI1 thru AI4 to the appropriate terminals on the back of the mounting cassette. Some examples of connections are shown in the figure. Select the AI signal ranges (0-20 mA/4-20 mA or 0-5 V/1-5 V) during configuration as explained in Section 5. Ensure that shields are connected to chassis ground by using either of the two lugs at the upper edge of the rear panel.

If a Pt-100 sensor or thermocouple is used without a transmitter, it must be connected directly to a DPR100 Temperature Input Module mounted on the module block. This disables the use of AI1 for any other signal.



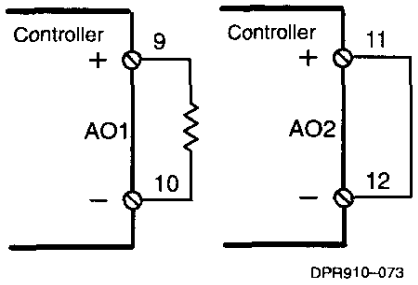
Notes:

1. Chassis ground
2. For voltage inputs the resistance jumper must be cut.

The DPR900 accepts signals from field-powered transmitters and can supply the required 24 Vdc to most remotely-powered transmitters. The figures show how to connect external sensors powered either by the controller or by an external source. The terminal block connector numbers for AI1 are shown, although the connection information applies to all analog inputs.

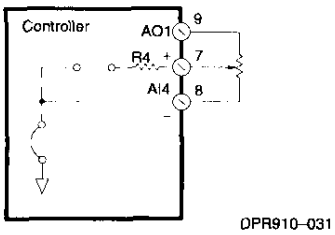
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2.8 Analog Output Signals



Two analog outputs are available on the DPR900 and both deliver a 4–20 mA/0–20 mA signal proportional to the 0–100% selected output signal. The controller PID output signal is always available at AO1. AO2 is a re-transmission of a user selected internal signal value.

Connect the analog output signal wires to terminals 9 and 10. Configure the output signal range using FCODE 116.

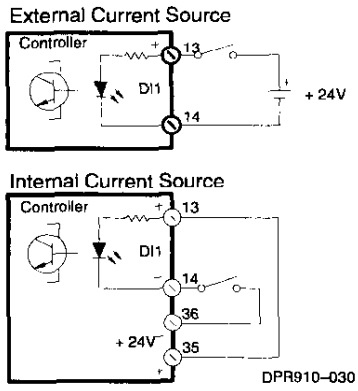


If an actuator position feedback potentiometer for a positioning motor output is required as shown in the figure. The potentiometer is connected to AO1 where the current input will be supplied.

Note

The negative input of any AI must be jumpered to supply ground when the potentiometer setup is used.

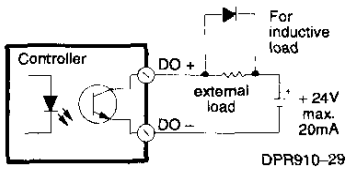
2.9 Digital Input Signals



All digital input signals (DI1 thru DI4) to the controller are isolated by opto-couplers. The input current source may be from the controller or from an external power supply as shown.

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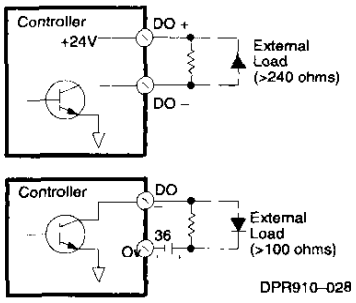
2.10 Digital Output Signals



Digital output signals DO1 and DO2 indicate the controller mode (auto/manual and remote/local) to an external destination. These two outputs are optically isolated. The figure shows an example of the connection for an external load with an external supply.

Note

If an inductive load such as a relay coil, is connected to a digital output, the coil must be shunted by a diode (1N4003 or similar). Observe proper polarity and wire the diode as close as possible to the coil.



Digital output signals (DO3 thru DO6) are open-collector transistor outputs as shown in the figure. The 24 V is directly connected to these outputs. However, each output can have its own separate supply.

Note

The 24 Vdc terminals can supply a maximum external load of 100 mA when powered from 110, 120, 220 or 240 Vac power sources. This supply is intended to power transmitters or relays connected to the discrete outputs. The 100 mA limit applies only to ac powered units. No internal load, including the two analog outputs, need be included in the 100 mA total.

Controllers powered from 24 Vdc sources have a current limit of 250 mA on each output.

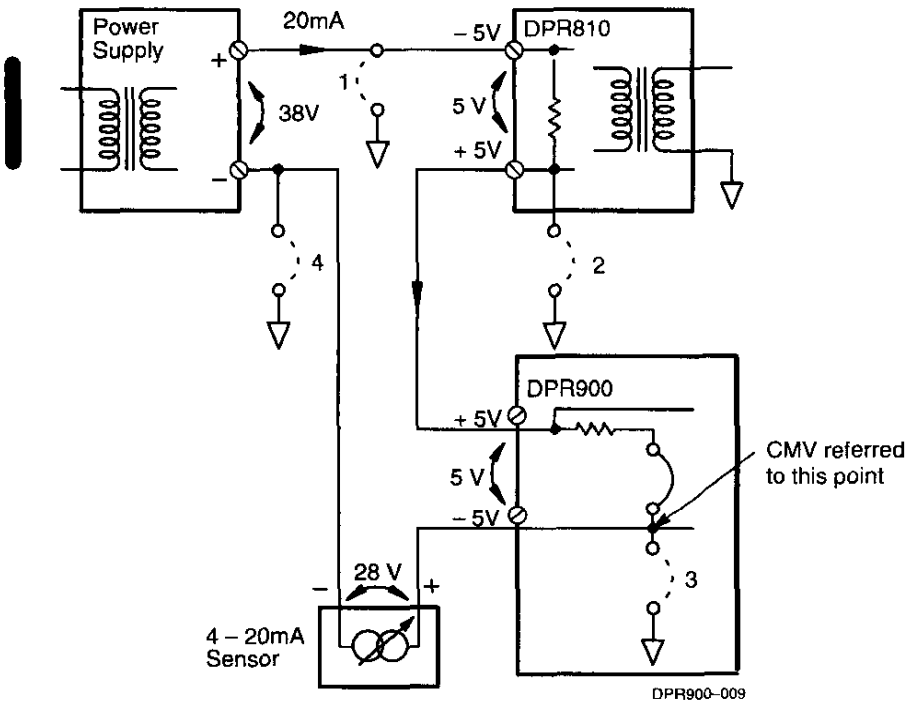
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2.11 Ground Loops and Common Mode Voltage

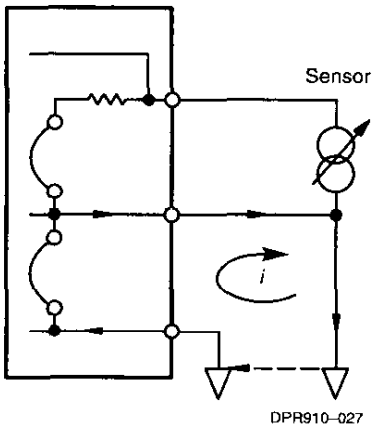
Ground loops occur when either the return wire or the sensor is grounded at the process line and Jumper 5 on the mounting cassette is intact. To correct this situation, the input should be connected as a differential input (Jumper 5 cut). The input then floats with respect to supply ground. The maximum common mode voltage (CMV) the controller should receive when wired in this manner is –5 to +17 volts taken from the input negative pole to supply ground. If these limits are expected to be exceeded, an isolation amplifier must be used.

Note

For each of the points 1 thru 4 in the following example, the table lists the maximum potentials and grounding possibilities.



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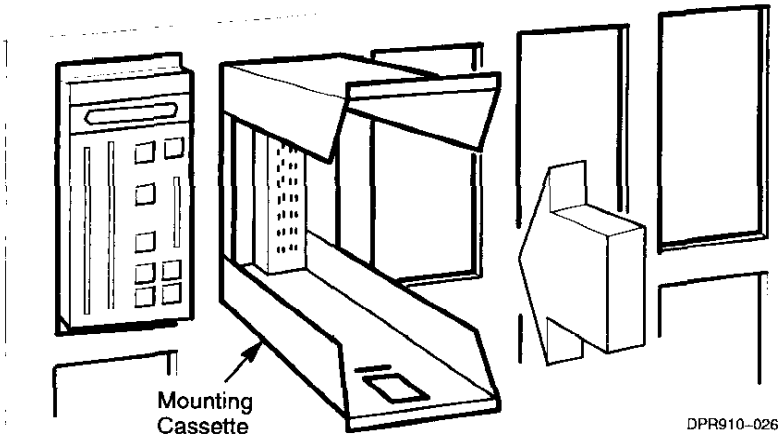
PT	Potential	Grounding Possibility
1	–10 V	Not permitted, Max CMV is –5 V
2	–5 V	Permitted
3	0 V	Permitted
4	+38 V	Not permitted, Max CMV is +24 V

The figure depicts a current loop example for a DPR900 controller. In this example, it is assumed that all units are isolated from the main power supply ground. This type of current loop must always be grounded at a single point so that a floating ground does not occur and introduce a large interference potential.

2.12 Installing the Mounting Cassette

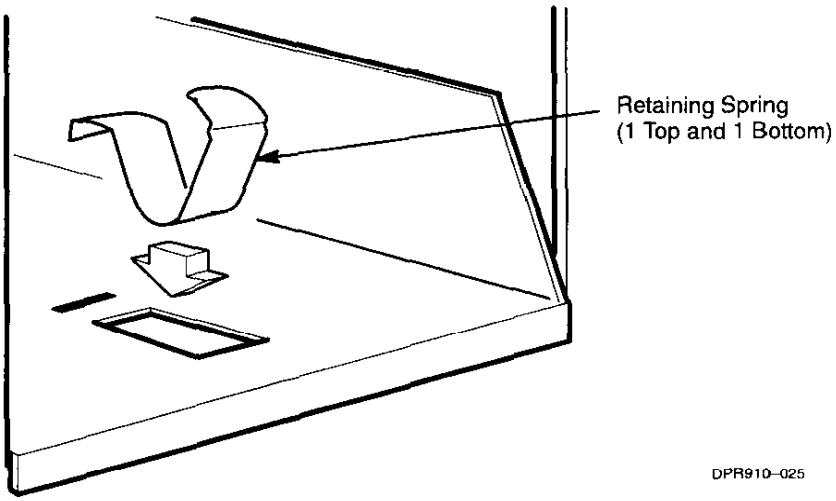
To fit the mounting cassette into the panel cutout:

- Position the mounting cassette (without the controller inserted) in the panel cutout as shown.



- Position the lower retaining spring as shown. Hold the mounting cassette in place with the thumb and use the index finger to press the spring down.
- Insert the upper retaining spring, as shown on the next page, to complete the installation.

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Note

In installations utilizing thicker panels, the retaining springs may not fit properly and can prevent insertion of the cassette. Contact your Fisher Representative for assistance.

When it is necessary to remove the mounting cassette, reverse the procedure. by first removing the upper retaining spring, then the lower one. Press the front edge of each spring back to remove it. In the case of a thick front panel, it may be necessary to use a screwdriver or other tool to apply enough pressure to remove the spring.

2.13 Installing the Controller

9147

DPR900-011

When the mounting cassette is firmly secured in the panel, install the controller by sliding it into the mounting cassette and pressing the unit firmly into place. When the mounting cassette is connected to its power source, and the DPR900 is fully seated in the mounting cassette, the front panel illuminates. The digital display will show the year and week (YYWW) of the controller firmware version. This display lasts 3 seconds before the controller begins operating to allow any transmitters connected to the controller to stabilize. The controller is now ready for use with the default configuration settings.

This manual supports software version numbers 8836 thru 9147. The functionality is limited to the software version currently loaded on the user's DPR900 Integral Controller.

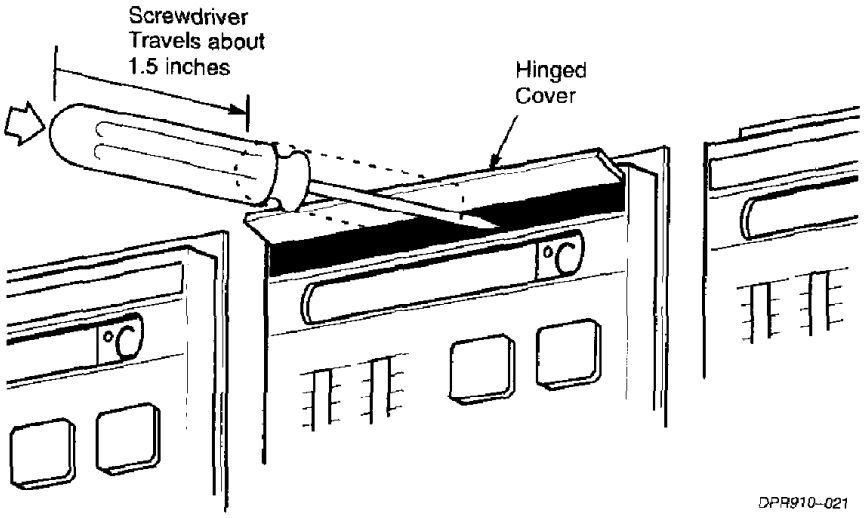
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2.14 Removing the Controller

To remove the controller:

- Raise the hinged cover and remove the configuration reference card.
- With a small screwdriver, push the sliding, red locking plate toward the back of the controller (the locking plate travels about 1.5 inches (38 *mm*) *before releasing*).
- Pull the controller out of the mounting cassette.

The hinged cover has two grooves that can accept a paper loop identification label. The configuration reference card is kept in the space directly behind this cover.



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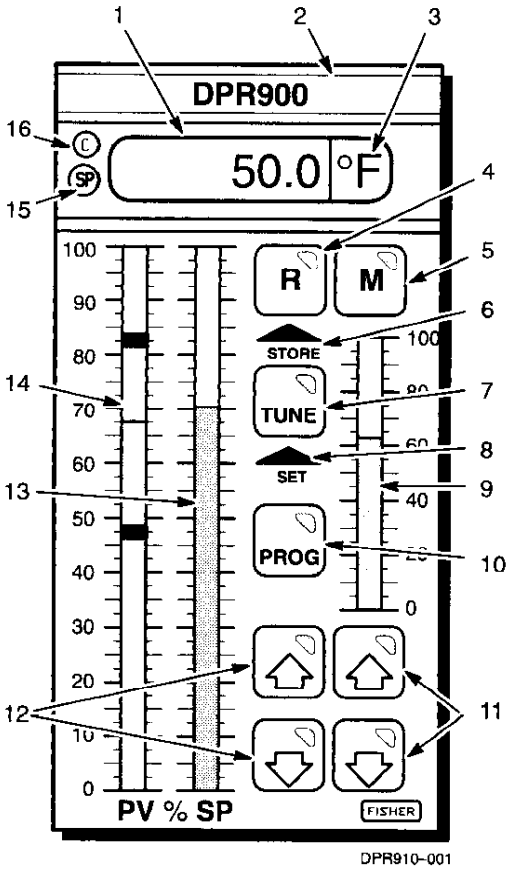
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3 Controls and Indicators

This section describes the DPR900 Integral Controller front panel controls and indicators, and gives general instructions for using them.

3.1 Controller Front Panel

The illustration shows the front panel controls and indicators, followed by a description of each.



DPR900 Front Panel Controls

1 – **Digital Display** — Shows process value and any alarm indications on five-character display. Can also display the SP value, configuration value, error code or selected analog input signal value.

2 – **Tag** — Paper label to identify controller.

3 – Engineering units display label.

4 – **R** Button — For switching between remote or extra internal setpoint value, computer or local control. The LED lights when the controller is in the remote mode.The button also functions as a **STORE** button during configuration. The **STORE** light (6) below this button lights when this alternate mode is active.

5 – **M** Button — Switching button for selecting Auto or Manual control. The LED lights when the controller is in the Manual mode. This button is also used to manually stop the Autotuner function.

6 – **STORE** Light — When this Red light is On, the alternate mode (STORE) of the **R** button (4) is active. When the light is flashing, it indicates that a configured value has been changed, and the user must store the new value before proceeding. Active only in Programming mode.

7 – **TUNE** Button — Used for starting and stopping the Autotuner function. The LED blinks while tuning is in progress. This button functions as a **SET** button during configuration. The Red **SET** light (8) below the **TUNE** button will be lit when this alternate mode is active.

8 – **SET** Light — When this Red light is On, the alternate mode (**SET**) fo the **TUNE** button (7) is active. This light will only be lit when the controller is in Programming mode.

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9 – Output signal vertical bargraph display (0-100%).

10 – **PROG** Button — Switching button for selecting the configuration mode. The red LED is lit during configuration. Selecting the configuration mode also lights the **SET** and **STORE** lights.

11 – **Increase/Decrease** Buttons — ON, Yellow, output signal can be stepped up/down one point at a time. These buttons are lit and active only when the controller is in the Manual mode.

12 – **Increase/Decrease** Buttons — ON, Green, setpoint and/or the configuration value can be stepped up/down one point at a time. These buttons are lit only when the controller is in the configuration mode or the setpoint is being manipulated.

13 – **SP Bargraph** — Analog setpoint (SP) vertical bargraph display.

14 – **PV Bargraph —** Analog process value (PV) vertical bargraph display. If alarm limits apply to the PV and are set in the controller, such limits are indicated by brighter bars on the display.

15 – **SP** — Setpoint light. Flashing or lit when setpoint is indicated on digital display.

16 – **C** — Lit when communication has been configured.

Press any of the four illuminated increase/decrease buttons, 11 or 12, to change the displayed value one increment (0.1 percent of measuring range). Holding the button down will continue to increase or decrease the value until the button is released.

The SP or configuration value buttons (12) also make changes in increments of 2.5% of measuring range. To increase/decrease this way, first press and hold the desired direction button, and then press the opposite button. The displayed value increases or decreases in increments of 2.5% of the full measuring range.

3.2 Digital Display

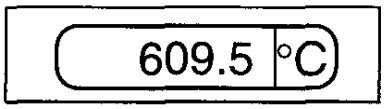
The digital display can indicate several different functions:

- Process value (normal display)
- Setpoint
- Alarms
- Fault codes
- Any analog input
- Function codes and associated parameter setting (only during configuration).

The following paragraphs address the displays available and give an example of each.

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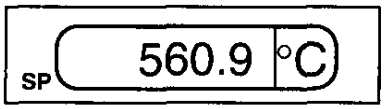
3.2.1 Process Value



DPR910-88

During normal operation the process value is displayed with the proper engineering unit.

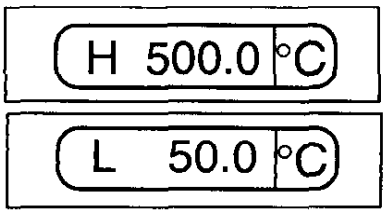
3.2.2 Setpoint



DPR910-87

When either of the green increase/decrease buttons is pressed, the setpoint is shown on the digital display. **SP** will be displayed on the left. After 5 seconds, the display returns to the process value.

3.2.3 Alarms



DPR910-86

A flashing **H** or **L** indicates that a high or low alarm limit has been reached. High-high and low-low alarm limits are indicated by a faster flashing rate. Alarm indications are displayed until the alarm condition is corrected.

3.2.4 Fault Codes

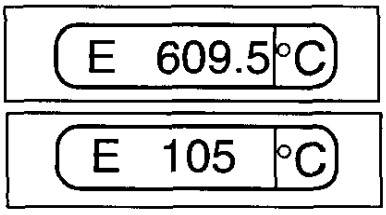
The types of fault codes displayed are:

- Wrong button error
- Warning
- Fatal error.



DPR910-856

An attempt to perform an improper operation by pressing one of the controller buttons (attempting to switch in computer control when it has not been configured), will cause **Error** to appear on the display for a few seconds.



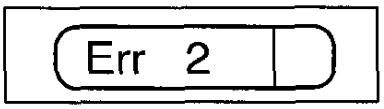
DPR910-89

A flashing **E** indicates a warning. Press the **PROG** button to check the type of warning condition present (refer to fault codes in Appendix A).

The fault can be acknowledged by pressing the **SET** button or saved by pressing the **PROG** button. If several faults are present, they are displayed by pressing the green increase/decrease buttons.

The display may be returned to the process value by pressing the **PROG** button twice.

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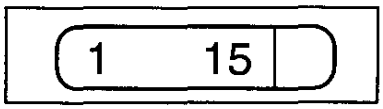
DPR910-084

Err n indicates the controller has a fatal error affecting the DPR910 function (refer to fault codes Appendix A). This type of fault cannot be acknowledged and must be referred to maintenance personnel.

3.2.5 Analog Inputs

Note

This function is intended for personnel involved with configuration.



DPR910-90

Any analog value can be shown as a percentage by selecting FCODE 143. The value is displayed as **X nnn** , where **X** is the input reference as shown in the following table, and **nnn** is the percentage value.

Ref Signal
1 AI1 in Engr Units
2 AI2 in Percent
3 AI3 in Percent
4 AI4 in Percent
5 AI5 in Percent

3.2.6 Display Priorities

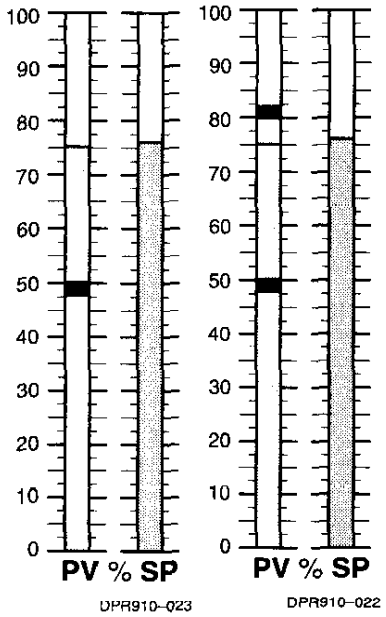
Because the left side of the digital display can show a number of different functions, the displays are given priorities.

The process value or the analog input/arithmetic block values are displayed for half the time and the rest of the time, the fault codes, autotuning phases and alarms are shown. When setpoint is shown, or the controller is in the Programming mode, no other functions are displayed.

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3.3 Bargraph Displays

3.3.1 Process Values



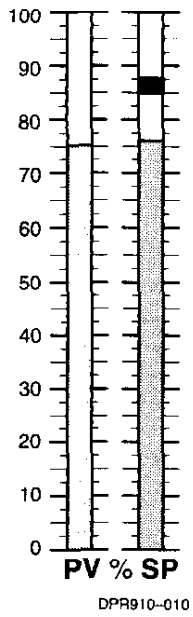
The process value bargraph displays the process value or any deviation alarm limits which may have been set. The two types of alarm limits are high/low and deviation alarms. Only one type may be present at a time.

High/low alarms are indicated by a brighter bar for each alarm setting. A maximum of two alarm points may be set. The upper and lower bright bars indicate the high-high and low-low alarms. If the alarm points are separated by less than 2.5%, only one bar is shown.

Deviation alarms are shown in the same manner as high/low alarms. The alarm limits however, are set relative to the setpoint and change when the setpoint changes. If the alarm limits are separated by less than 2.5%, only one bar is shown.

The display at the left indicates a PV low setpoint deviation alarm of 25%. The display at the right indicates PV alarms set at 82 and 50%.

3.3.2 Setpoint



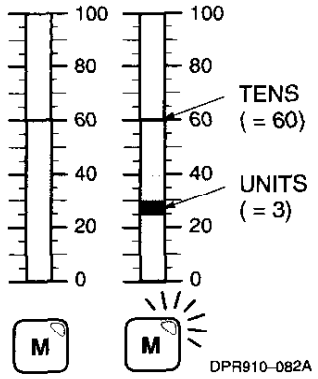
The setpoint bargraph normally displays the setpoint value. If the point ramp is in operation, the setpoint final value is shown as a brighter bar during the ramp operation period.

In the display shown, the final operator set ramped setpoint is 87%, while the current setpoint is 75%.

3.3.3 Output Signal

The reading of the output signal display is dependent on the mode of operation selected.

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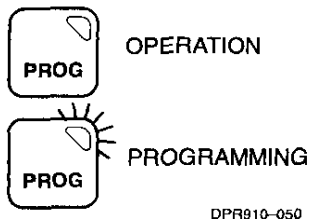


In Auto (Remote) control, the output signal bargraph displays a dull control signal as 0-100% or 100-0% (determined by the configuration). The resolution in this mode is 5%. For actuators with position feedback, the actuator position can be indicated on the bargraph. In the example on the left, the indicated output signal is 60%.

In the Manual mode, the figure on the right indicates the manually adjustable output signal with increased resolution to 0.5%. The normal brightness bargraph (graduated in tens as indicated) shows 60% and the brighter bar (graduated on the left side of the scale in units 0-10) shows 3%. This combination represents a manually adjusted output signal level of 63%.

3.4 Operating Modes

3.4.1 Operation/Programming



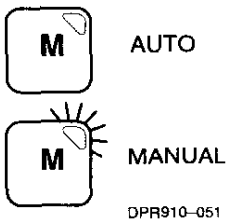
In the Operation mode, the front panel buttons perform their normal functions for operation and setting of the setpoint and output signal values.

In the Programming mode, the function buttons have alternative functions for programming (and alteration of the controller type during cascade control). The Programming mode should only be used by configuration personnel.

Note

The DPR900 configuration settings (FCODE 1-46) can be altered during operation by the Programming mode. Changes to FCODE's higher than 100 require the controller to be switched out by setting it to Manual.

3.4.2 Auto/Manual



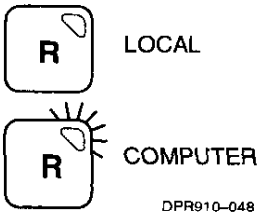
In Auto control, the PID control system is in operation. In Manual control, the PID control system is switched out and the input signal can be controlled directly from the controller's increase/decrease buttons. Normal operation is with the DPR900 in the Auto mode.

Switching between Auto and Manual can be done by:

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- pressing the **M** button
- an external signal connected to one of the controller's digital inputs
- a signal from a remote computer.

3.4.3 Computer/Local Control



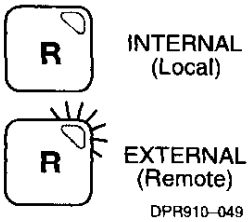
In Computer control, some of the controller's functions are controlled from a remote computer. In Local control, the front panel controls are operational.

If the controller is not connected to a remote computer, Local is the only mode used. If the DPR900 is connected to a remote computer, Computer mode is the normal mode of operation.

Switching between Computer and Local control can be done by:

- pressing the **C/L** button
- an external signal connected to one of the controller's digital inputs

3.4.4 Remote/Local Setpoint



In Remote setpoint, the controller setpoint is taken from an external source, usually a signal connected to one of the controller's analog inputs. In Local setpoint, the setpoint is set by the controller's increase/decrease buttons.

If the controller is not supplied with a remote setpoint signal, Local is the only mode used. Remote setpoint is blocked by configuration.

Switching between Remote and Local setpoint can be done by:

- pressing the **R** button
- an external signal connected to one of the controller's digital inputs.

3.4.5 Normal Operating Modes

- Single controller operation — Auto, Local, Extra Internal setpoint
- Controller with external setpoint signal operation — Auto, Local, Remote setpoint
- Computer control operation — Auto, Local, Remote setpoint (R button has no function).

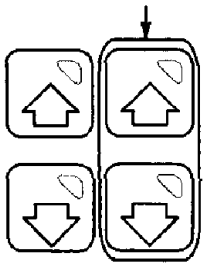
3.5 Manual Control of Output Signal

1. Switch to Manual mode by pressing the **M** button. The yellow increase/decrease buttons light.

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Note

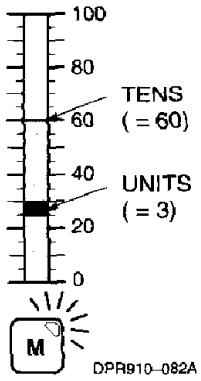
The **M** button also lights to indicate Manual when the output signal can be controlled by remote signals. The yellow increase/decrease buttons are disabled.



DPR910-047

The output signal indicator changes to show the output signal with 0.5% resolution, compared to the normal 5%. This is done by combining the normal bargraph with a brighter bar. The normally bright column indicates the ten units, and the brighter bar single units.

Example:



DPR910-082A

The normal brightness bargraph shows 60% and the brighter bar shows 3%. This combination represents an output signal level of 63%.

If the increase/decrease button is held until the single unit bar goes beyond 0 or 100%, the bargraph indication will revert to its normal 5% resolution and the stepping speed will increase to 5% per step.

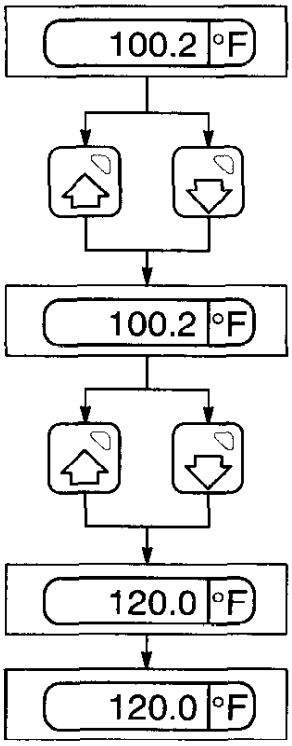
2. Use the yellow increase/decrease buttons to step to the required output signal value. if the output signal has its limits applied for Manual setting, the bargraph indication will stop at the limit.
3. Return to the Auto mode by pressing the **M** button.

3.6 Establishing a Setpoint

3.6.1 Local Setpoint (LSP)

The controller must be in the local mode to enter a local setpoint (LSP). The LED in the **R** should be off. If the LED is lit, pressing the **R** button switches it back to the the local control mode.

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DPR900-012

To set the value:

1. Momentarily press and release either the setpoint increase or decrease button (12). The green LED in both buttons lights, indicating that the setpoint value may be changed.

Note

The controller must be in the proper mode or the local setpoint value will not take effect. Ensure that the remote button LED is not lit and the green setpoint increase/decrease button LED's are lit before attempting to alter the setpoint. During local setpoint entry the digital display shows the local setpoint value. The **SP** light to the left of the display illuminates.

2. Set the new setpoint value using the increase/decrease buttons (12).

3. The controller displays the new setpoint value in the digital display for about seven seconds after release of the button, then reverts to the configured display value.

If setpoint limits are configured in the controller, the controller does not accept values outside such limits. The value shown on the digital display will remain constant once a configured limit has been reached. The SP bargraph also shows a full scale reading, 0-100% of the configured setpoint range.

3.6.2 Remote Setpoint/Extra Internal Setpoint/Computer Setpoint

The **R** button enables the remote setpoint value (RSP), extra internal setpoint value (ESP), or the computer-generated setpoint value (CSP). The green LED in this button lights to indicate Remote control. The RSP value may be a setpoint value from another device, an ESP value stored in the controller, or a CSP sent via the RS-485 communications link.

With the controller in the remote mode, display the actual RSP value on the digital display by momentarily pressing either the setpoint increase or decrease button. The **R** button LED remains lit as a reminder that the displayed setpoint value cannot be altered manually.

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Note

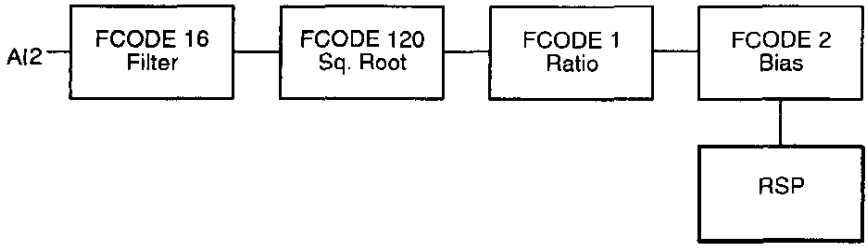
The Local setpoint cannot be changed while the controller is in the remote mode. If the user attempts to change the setpoint value in the remote mode, the controller flashes **ERROR**.

3.6.3 Switching Between Remote and Local Setpoints

Switch from the RSP, CSP or ESP mode to the LSP mode by using an external digital signal or switch manually, using the **R** (remote) button. Manual switching via the **R** (remote) button always overrides the digital signal or RS-485 signal. Remote/Local switching occurs only if the controller is in the Remote mode.

3.7 Changing Configured Values: Ratio, Bias and Extra Internal Setpoint

The RSP input value received at AI2 can be configured to be modified before being used as the controller's RSP.



DPR900-013

Use FCODE 1 and 2 to adjust the ratio and bias values used in conjunction with the RSP or the internal calculation functions (FCODE 107 and 108). Both values reside within the controller configuration parameters; therefore the user must enter the configuration mode to change these values. Press the **PROG** button to enter the configuration program. Then use the **TUNE** button to advance to the appropriate FCODE. After each FCODE number, the controller indicates the current value stored. Use the increase/decrease buttons (11) to change the value. Press the flashing **R** button to save the new value.

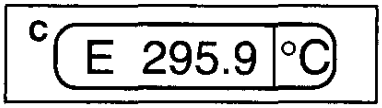
Function	Parameter	FCODE	Range	Unit	Increment	Default Value	Selected Value
Ratio	Gain	1	0–9.99	–	0.01	1	
Bias	Zero level offset	2	+/– M	any	1 digit	0	

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The controller can also be configured to switch to a preconfigured ESP value upon certain discrete switching and fault situations. Use FCODE 12 to configure this value.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Extra Internal SP	Setpoint value	12	Meas.	any	1 digit	0	

3.8 Computer Control



DPR900-017

Computer control is available when the red **C** in the upper left of the digital display is lit. Use the **R** button to select computer control when it is available. When computer control is active the **R** button green LED is on. When the controller is in local control mode, the computer can read values from the controller. Under computer control, two discrete input signals control the setpoint value, the output signal or accept control signal changes via the RS-485 communications link.

The following conditions apply to computer control:

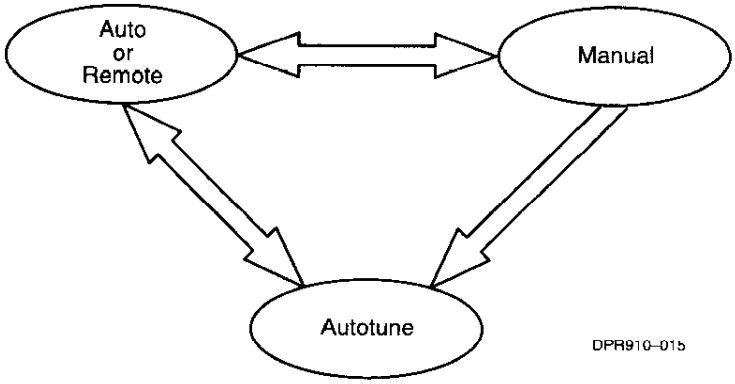
- If the controller has been switched to computer control (**R** button lit), switching between computer mode and local mode is accomplished via an external discrete signal.
- If the external discrete switching signal is deactivated, the controller switches to local mode and the green LED in the **R** button remains lit.
- The controller then uses the LSP, RSP, or ESP, depending on which options are used for FCODE 141 and 142.

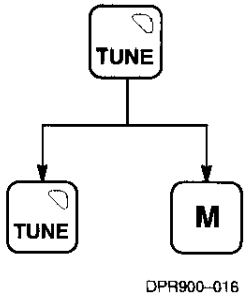
3.9 Autotuner Operation

The autotuner is a utility for automatic setting of PID parameters. This function can also be used with non-linear processes when used with gain. Before tuning is started, ensure that the process is stable at a position close to the required working point, i.e. SP = PV.

With the controller in either Remote, Auto or Manual, use the **TUNE** button to start the autotuner. The green LED in the button blinks while the autotune mode is active. For slow processes the autotune function may take as long as 32 hours, the maximum time the autotuner will run. The controller returns automatically to auto mode when the tuning procedure is complete.

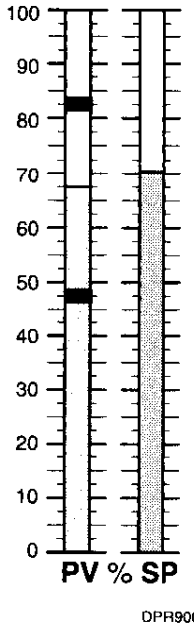
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To abort autotuning, press either the **TUNE** button a second time or press the **M** button. Pressing the **TUNE** button returns the controller to either Auto or Remote mode. The controller always returns to the auto/remote mode from the autotune function. Pressing the **M** button stops the autotune function and switches the controller to the manual mode. If the user starts the autotuner when the controller is in the manual mode, the controller automatically switches to the auto mode, completes the tuning cycle, and returns to the auto mode when tuning is complete.

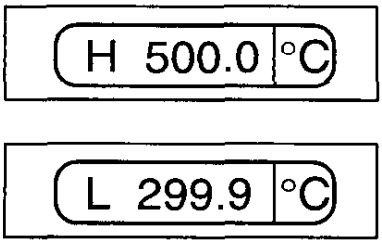
3.10 Alarm Indications



The PV bargraph display shows the currently set alarm limits as brighter bars. Up to two alarms can be displayed. If a limit is exceeded, a flashing **L** or **H** is shown in the left position of the digital display. A discrete output may also indicate any alarms that have been configured.

In the illustration, a high alarm is set at 82.5 and a low alarm is set at 47.5.

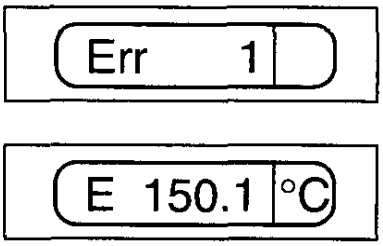
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DPR900-019

The **H** and the **L** in the digital display flash faster (2X) than the normal high and low alarm indications to show that the high-high or low-low limits have been exceeded.

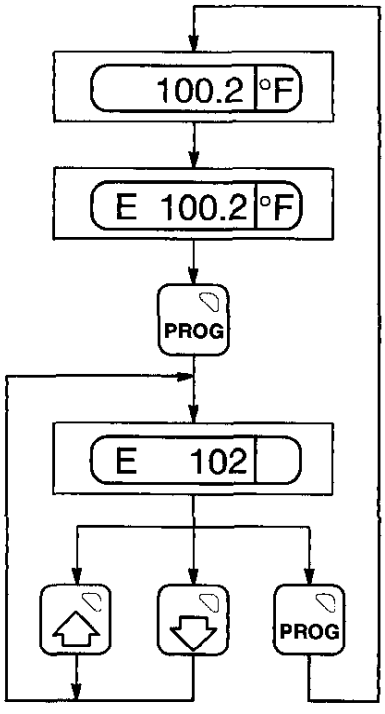
3.11 Error Indications



DPR900-020

If the controller detects a fault during continuous automatic self-testing, **Err n**, or a flashing **E** appears in the left character position of the digital display. When this happens the user can press the **PROG** button to display the error code. Refer to Fault Codes, Appendix A.

If more than one error code occurs, each code may be called up in turn by pressing the setpoint/configuration increase/decrease buttons while in the Programming mode.



DPR900-021

There are two ways of exiting the error indication mode:

1. Pressing the **PROG** button once more puts the controller in the configuration mode. The error codes remain stored in the controller and may be examined again, if necessary. Pressing the **PROG** button a third time takes the controller out of the configuration mode and returns the displays to their normal states.

2. Pressing the **TUNE** button also switches the controller to the configuration mode, but erases the error codes. Pressing the **PROG** button again returns the controller to the normal display.

Error codes lower than 101 indicate fatal errors which seriously effect the controller function

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Note

Clearing an error code upon entering the configuration mode, may not clear the code due to required configuration changes. Upon completion of these changes and exiting the programming mode, the original error code may still be observed. Re-enter the configuration mode a second time to clear the error.

3.12 Bumpless Transfer

Switching between Auto and Manual is bumpless (unchanged) in both directions. This does not apply if the DPR900 is used as a simple proportional (P) controller. In a P controller, the output signal in Auto is always 50% when the process value is equal to the setpoint value.

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4 Function Codes

4.1 Selecting and Changing Function Codes

Function code settings (FCODE's) determine the internal values for:

- Tuning Parameters
- Configurable Options

These values are determined by configuration and should be entered by the user in the FCODE tables.

Setting and checking of these FCODE's in the DPR900 can be carried out at one time with the instrument in the **Programming** mode.

The instrument need not be connected to sensors and actuators for FCODE setting. If, however, setting is to be performed without any connection to AO1, Terminal 9 and 10 must be short-circuited to prevent an open-circuit error message (ERR 4) from being initiated.

All FCODE's can be altered during operation, however, to change configuration options, the controller must be in the Manual mode.

4.2 Setting Principle



Note

To set or alter FCODE 101 or higher, the DPR900 must be set to the Manual mode. This ensures that the output signal is under local control and cannot be provided with an unwanted output value.

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Note

FCODE 24, 25 and 26 must be set for the appropriate measuring ranges before any other FCODE is selected. This is because many other parameters use the measuring range limits as reference. If the measuring range is altered, related limits and alarms are automatically adjusted to suit the new range. (This does not apply to cases where the DPR100 Temperature Input Module is used since its measuring range is set automatically with FCODE 118.

Tuning parameter setting and configuration function selection take place in the configuration mode.

Press the **PROG** button to enter the configuration mode. Pressing the **PROG** button illuminates the red light in the button and the red **STORE** and **SET** lights. These lights indicate that the secondary functions of the **R** and **TUNE** buttons are active.

If the controller is set for authorization check (FCODE 139), the configuration values may be read only when the **STORE** light is Off.

Most configurable controller functions and tuning parameters can be changed while the controller is operating. For safety, vital controller functions cannot be altered during operation. The controller must be in the Manual mode to change any functions with FCODE's greater than 100. When changes are made during configuration, either the **SET** or **STORE** light flashes, indicating the appropriate action to be taken.

A complete list of FCODE's is provided in Section 5.

4.2.1 Tuning Parameter/Configuration Changes



Note

The authorization check can be temporarily disabled (without changing FCODE 139) by pressing the **R** button immediately after pressing the **PROG** button in Step 1.

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Change tuning parameters according to the following procedure:

Step	Button	Display	Action
1	PROG	PROG or E107	Select program mode; **PROG** button LED starts flashing. If an error code is present (e.g. E107), it appears in the digital display. Call up any additional error codes by pressing either green increase or decrease buttons.  Leave the error code mode without clearing the error by pressing,  or Press **TUNE** button to erase the error codes.
2		FC 1	Authorization check is configured. The configured values may only be viewed, not changed (Steps 2 and 3). Steps 4 and 5 change values only when authorization check is not configured.
3		FC n	Step forward (using the green increase button) or backward (using the green decrease button) to the desired function code. The **SET** button flashes.
4		50.50	Push the **SET** button to view the current parameter value for the selected function code.
5		55.50	Using the green increase and decrease buttons, set the new value in the digital display. The **STORE** button starts to flash, indicating that the changed value must be stored to take effect.
6		FC n	Press the **STORE** button to store the new value; the **PROG** button starts to flash, and the digital display shows the next function code.

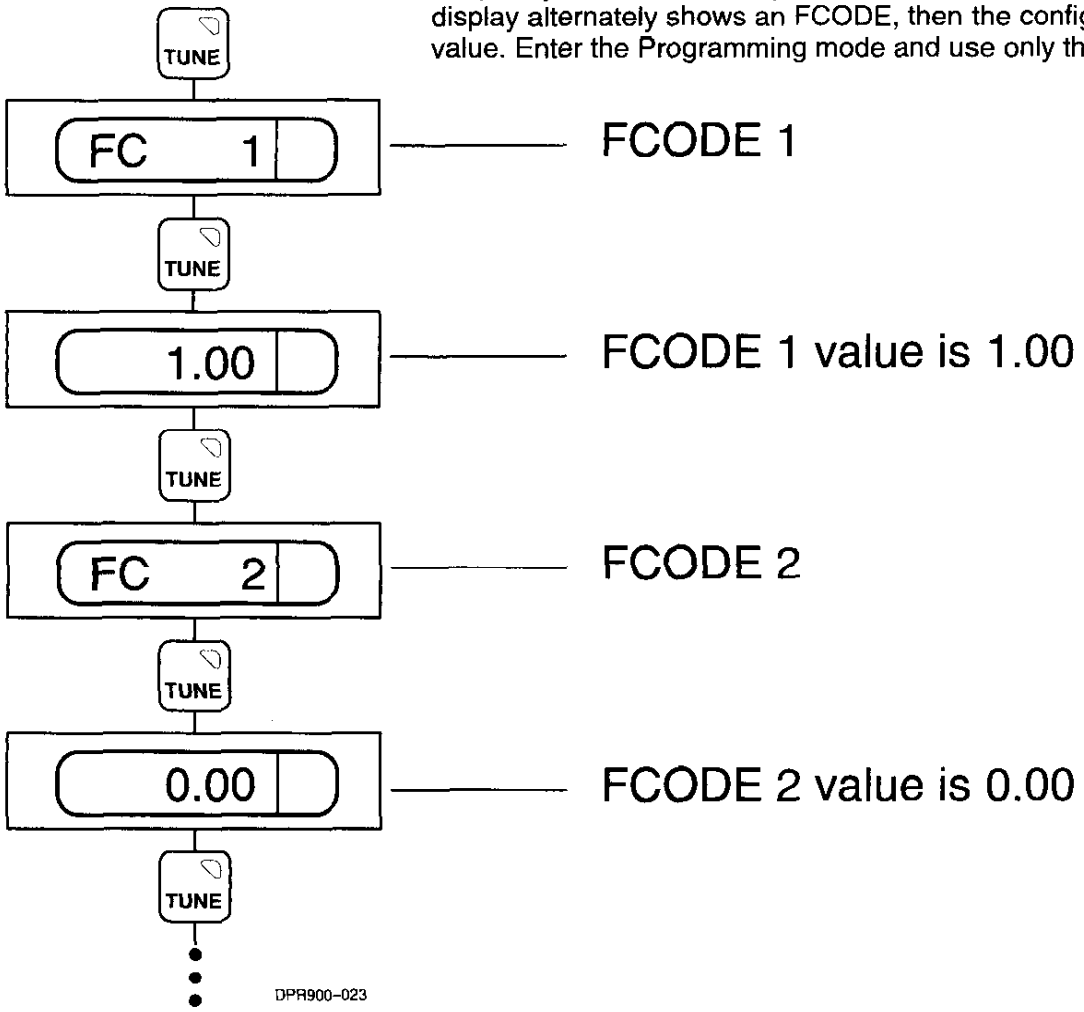
Call other function codes by using the green increase/decrease buttons, then complete Steps 3 thru 6 to change the parameters.

7		27.3	Press the flashing **PROG** button; the controller leaves the programming mode, and the display shows the configured display value.

DPR900-022

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To quickly view the current parameters, use the **TUNE** button. The display alternately shows an FCODE, then the configured parameter value. Enter the Programming mode and use only the **TUNE** button.



4.2.2 Function Selection

Function code 101 thru 145 are used in configuring the controller. The controller must always be in the manual mode before any FCODE can be selected. The programming procedure is identical to that described above for tuning parameter changes, but instead of parameters, option numbers (0–1–2–3–4–5) are set. Each number represents a particular function.

Note

When an FCODE is selected for display, the function (number) currently in force is shown.

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5 Configuration

5.1 General

Configuration of the controller follows three stages:

- Definition of the control problem.
- Selection of the inputs and outputs.
- Selection of the required functions (FCODE's).

The controller is delivered with a basic configuration, permitting it to be used as a conventional PID controller. All function codes have been assigned the default values listed in the FCODE tables.

5.2 Definition of the Control Problem

Before beginning to look at the controller configuration, the control application must be carefully defined.

- Which process is to be controlled?
- What is to be achieved?
- Are alarm function needed?
- Which input and output signals (external components) are available and what are their signal ranges?

A control system diagram should preferably be drawn up in the form of a block diagram which will answer the questions posed above.

5.3 Selection of Inputs and Outputs

The controller has:

- Four analog inputs: AI1, AI2, AI3, AI4
- Two analog outputs: AO1, AO2
- Four digital inputs: DI1, DI2, DI3, DI4
- Two optically-isolated digital outputs for mode indication: 24V/20 mA: DO1, DO2
- Four digital outputs for alarms and discrete output control: DO3, DO4, DO5, DO6

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When the controller configuration is fully planned, enter the configuration options and the tuning parameters. Press the **PROG** button to access the configuration mode. The controller is designed to prompt the user for easy configuration entry. Use the **TUNE**, and green increase/decrease buttons, to call the appropriate FCODE's, select and enter the functions, and set the tuning parameters.



Note

Set the measuring range and fix the position of the decimal point in the digital display, using FCODE 24, 25 and 26 **before** changing any parameter. Many parameters use the measuring range limits as a reference. If the measuring range is altered, the limits and alarms that refer to the measuring range adjust automatically. This eliminates the need for further adjustments.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Measuring Range	Minimum value	24	± 9999	any	1 digit	Min	
	Maximum value	25	± 9999	any	1 digit	Max	
	Decimal point	26	1 – 4	–	1	1	

5.4 Input Signal Assignment

Some input and output signals have definite functions; (e.g. AI1 is always the process value [PV] input). Other signals, such as AI3, AI4 and DI1, DI2 can have any **one** of the assignments shown. The user assigns one of these options to the signal line, then chooses the valid signal range and type of input that indicates the active discrete signal.

Signal	Assignment	Signal	Assignment
AI1	Process value	DI1	For switching between:
AI2	Remote setpoint		o Auto/Manual
AI3	External analog signal for:		o RSP value to ESP value
	o Secondary process value monitoring sensor		o Computer to local control
			o Forced output signal
	o 3-state control		o Output tracking
	o Min/Max selection	DI2	For switching between:
	o Arithmetic functions		o Auto/Manual
	o Feedforward signal		o RSP value to internal alternate signal
	o Gain scheduling signal		
AI4	External analog signal for:		o Computer to local control
	o Showing control position		o Forced output signal
	o 3-state control		o Output tracking
	o Feedforward signal	DI3	Digital control of output signal
	o Gain scheduling signal		DDC alternatively setpoint SPC
	o Output tracking	DI4	Digital control of output signal
	o Min/Max selector		

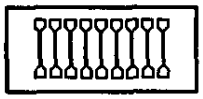
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It is not possible to assign two different functions to one input. If the user attempts to assign a second function to an input that has already been assigned, **ERROR** flashes on the digital display. When this happens, the user should assign the function to an unassigned input. Alternatively, the user may reconfigure the input signal line by removing the original function.

5.4.1 Analog Inputs (AI)

5.4.1.1 Current or Voltage Inputs

To use an input as a current input, the appropriate jumper must be intact.



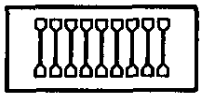
DPR900-58a

Jumper 1 = AI1, Jumper 2 = AI2, Jumper 3 = AI3 and Jumper 4 = AI4.

These jumpers connect 250 ohm resistors in parallel with each input. To use an input for voltage, the appropriate jumper must be cut.

5.4.1.2 Fixed or Floating References (Return Lines)

To provide an input with a fixed reference the appropriate jumper must intact.



DPR900-58a

Jumper 5 = AI1, Jumper 6 = AI2, Jumper 7 = AI3 and Jumper 8 = AI4

The jumpers set the common-mode voltage for each input to the range –5V to +17V. The 5V signal voltage is added to this, giving a total or 17 +5 = 22V, which can vary by +/– 10%. In order to obtain a floating reference the appropriate jumper must be cut.

5.4.1.3 Low-Pass Filter

A first-order, low-pass filter for any of the analog inputs, may be enabled if required. To enable a filter, use FCODE 15 thru 18 to select the filter time constant. Set a time constant using the green increase/decrease buttons on the front panel.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Analog Input Filter	Time constant AI1	15	0–9999	s	1s	0	
	Time constant AI2	16	0–9999	s	1s	0	
	Time constant AI3	17	0–9999	s	1s	0	
	Time constant AI4	18	0–9999	s	1s	0	

5.4.1.4 Measuring Range

The first parameters to be determined are those for the analog input measuring range, and the number of decimal places to be shown on the digital display. Use engineering units, such as ft^3/hr (m^3/hr), gals/min (liters/min, °F (°C), etc.

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The measuring range is set using FCODE 1 to 3.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Measuring range	Min value Max value No. of	24 25 26	± 9999 ± 9999 0–4	any any –	1 digit 1 digit 1	0.0 100.0 1	



Note

The parameters should be inserted in the order FCODE 26 thru 24 (number of decimals first) to obtain the correct values.

Example:

A level controller is to regulate between 0.39 and 1.18 inches (10 and 30 mm). The sensor supplies 0 mA for 0 inches (0 mm), and 20 mA fpr 1.97 inches (50 mm). Set FCODE 3 to 1, FCODE 1 to 0.0 and FCODE 2 to 500.

5.4.1.5 Signal Range Selection

Use FCODE 119 thru 122 to select the appropriate electrical signal range and linearization option for each of the four analog inputs, AI1 thru AI4. Select one of the options. To enable the square root option for AI1, use FCODE 107.

Function	FCODE	Option	No	Default Value	Selected Value
AI1	119	4–20mA, 1–5V	1	1	
		0–20mA, 0–5V	2		
AI2	120	4–20mA, 1–5V	1	1	
		0–20mA, 0–5V	2		
		4–20mA, 1–5V, square root	3		
		0–20mA, 0–5V, square root	4		
AI3	121	4–20mA, 1–5V	1	1	
		0–20mA, 0–5V	2		
		4–20mA, 1–5V, square root	3		
		0–20mA, 0–5V, square root	4		
AI4	122	4–20mA, 1–5V	1	1	
		0–20mA, 0–5V	2		
		4–20mA, 1–5V, square root	3		
		0–20mA, 0–5V, square root	4		

Function	FCODE	Option	No	Default Value	Selected Value
Calculation of Process Variable	107	Disabled	0	0	
		Square root	1		
		R = (AI1 x Ratio) + Bias	2		
		R = AI1 + (AI3 x Ratio) + Bias	3		
		R = (AI1 x Ratio) + AI3 + Bias	4		
		R = (AI1 x AI3)(Ratio) + Bias	5		

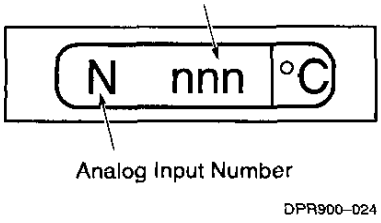
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Note

When supplying a voltage signal on any of the analog input signal terminals, ensure that the appropriate jumper is cut at the back of the mounting cassette. Cutting the jumper disconnects the 250 ohm shunt resistor.

Use FCODE 143, to display any of the analog inputs other than the default (process value, input signal AI1). When the default value is used in conjunction with FCODE 106, the value indicated is in rescaled units.

Function	FCODE	Option	No	Default Value	Selected Value
Normal Display	143	Display process value	1	1	
		Display AI1, in percent	2		
		Display AI2, in percent	3		
		Display AI3, in percent	4		
		Display AI4, in percent	5		



The other display options show the analog input values as a percentage, after filtering (if selected) and linearization (if selected). When the display shows an analog input other than the process value, the input signal number appears to the left of the percentage value.

The user may configure input signals AI3 or AI4 as reference signals for the gain scheduling function. When one of these inputs is used as a reference signal, the user may wish to configure the selected reference signal to appear on the digital display by setting the value of FCODE 143 to option 4 or 5.

5.4.2 Analog Inputs Using DPR100 Series Temperature Input Module

The DPR900 accepts either a 4–20 or 0–20 mA signal from a temperature transmitter, direct RTD or thermocouple temperature-sensor input via a DPR100 Temperature Input Module. Any temperature module connected to the controller sends its input signal via the AI1 terminals, disabling AI1 for any other input.

When a temperature input module is present, ensure that Jumper 1 is cut and Jumper 5 is intact. Set FCODE 119 to Option 2 (AI1 signal range 0–20 mA). Also select the appropriate option for FCODE 118.

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Function	FCODE	Option	No	Default Value	Selected Value
AI1 Signal Range	119	4–20mA, 1–5V 0–20mA, 0–5V	1 2	1	



Note

Do not use FCODE 118, unless a DPR100 Temperature Input Module is being used. Also, ensure that the appropriate temperature range is used (°C or °F).

Function	FCODE	Option	No	Default Value	Selected Value
Temperature Input AI1	118	Disabled 100 ohm Platinum RTD, °C Type J Thermocouple, °C Type K Thermocouple, °C Type R Thermocouple, °C Type S Thermocouple ,°C 100 ohm Platinum RTD, °F Type J Thermocouple, °F Type K Thermocouple, °F Type R Thermocouple, °F Type S Thermocouple, °F	0 1 2 3 4 5 6 7 8 9 10	0	



Note

Ensure that the square-root extraction function (FCODE 107) is set to 0 (disabled), if a temperature input module is connected to the controller.



Note

When using the PV monitoring function, the second temperature transmitter (connected to AI3) must have the same measuring range and engineering unit (EU) values as the transmitter or sensor connected to AI1.

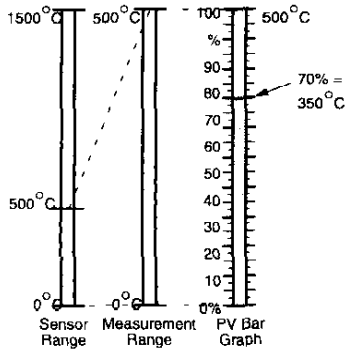
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Note

The resolution is the measuring range shown in the following table, divided by 4000. The resolution will not be improved by using limits other than those in the table. Even though the stated resolution is fractional, the digital display only shows degrees in whole numbers.

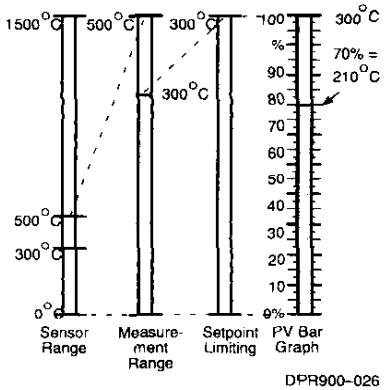
Module Type	Degrees F		Degrees C	
	Range	Resolution	Range	Resolution
100 ohm RTD	–58 – 1112	0.29 degrees	–50 – 600	0.16 degrees
Type J	32 – 1472	0.36 degrees	0 – 800	0.20 degrees
Type K	32 – 2372	0.58 degrees	0 – 1300	0.32 degrees
Type R	32 – 2732	0.67 degrees	0 – 1500	0.37 degrees
Type S	32 – 2732	0.67 degrees	0 – 1500	0.37 degrees

Examples:



DPR900-025

A Type R Thermocouple sensor with a DPR100 Temperature Input Module has a sensing range of 0 to 1500°C. The measuring range is 0 to 500°C, set with FCODE 24 thru 26. The controller sees only that portion of the sensor signal lying within the defined measuring range. The digital display now shows temperature between 0 and 500°C.



DPR900-026

The displays can now be further limited by using the setpoint limiting function to limit the maximum setpoint range (FCODE 106, 9, 10) to 0–300°C. If setpoint limits of 0°C and 300°C are entered, the digital display indicates exact temperatures between 0 and 500°C, but the bargraph indicates 0°C at 0% abd 300°C at 100%. At 70% on the bargraph, the temperature is 210°C.

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5.4.3 Discrete Inputs (DI)

Use FCODE 125 thru 128 to assign discrete input functions to DI1 thru DI4. The controller recognizes a discrete input signal as **active** when voltage is present at the input.

Function	FCODE	Option	No	Default Value	Selected Value
DI1	125	Voltage present at input	1	1	
		No voltage present at input	2		
DI2	126	Voltage present at input	1	1	
		No voltage present at input	2		
DI3	127	Voltage present at input	1	1	
		No voltage present at input	2		
DI4	128	Voltage present at input	1	1	
		No voltage present at input	2		

5.4.4 External Discrete Signal Switching (A/M and R/L) and Signal Tracking

Discrete signals DI1 and DI2 are used to switch modes from auto to manual and/or from remote to local. While the discrete input allows changing of states, the initial state must have been enabled from the front panel (i.e. to go from local to remote, the **R** button must initially be set from the front panel). The discrete cannot be used to switch the controller to the remote mode initially. Mode switching is on the trailing edge of the discrete signal. Only the **M** button lights when the controller is switched remotely. The yellow increase/decrease buttons do not light, and do not function unless the **M** button has been pressed. This activates the local/manual control state.

Setpoint tracking via FCODE 140 **does not** occur when remote switching selects the manual mode. This prevents the accidental reset of the setpoint in cases such as cascade control.

This function also controls the activation of output tracking if it has been selected as active in the manual mode. To control the signal manually, even after remote switching to manual, the user must press the **M** button to activate the yellow increase/decrease buttons.

The controller cannot be switched into the auto mode as long as the discrete signal remains active. DO1 (Auto/Manual status) is activated when the controller is in manual mode and remains active until the controller is in auto mode.

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Note

The user must press the **M** button to control the output signal manually even though a discrete input may have switched the controller to manual mode. The manual mode yellow increase /decrease buttons only light after pressing the **M** button to indicate that manual control is possible. When output signals are remotely controlled, pressing the **M** button does not light these two buttons, indicating that they are not locally adjustable.

Use FCODE 110 to select the function and the input signal line, DI1 or DI2.

Function	FCODE	Option	No	Default Value	Selected Value
External A/M Track Switching	110	Disabled	0	0	
		Auto/Manual/Track switching via DI1	1		
		Auto/Manual/Track switching via DI2	2		

Use FCODE 111 to select the external remote/local switching option.

Function	FCODE	Option	No	Default Value	Selected Value
External R/L Switching	111	Disabled	0	0	
		Remote/Local switching via DI1	1		
		Remote/Local switching via DI2	2		

Use FCODE 112 to select the option for forced control of the output signal via DI1 or DI2.

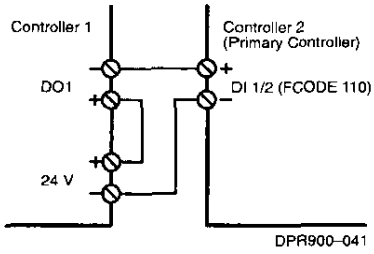
Function	FCODE	Option	No	Default Value	Selected Value
Forced Output Control	112	Disabled	0	0	
		DI1, applies in Auto/Manual	1		
		DI2, applies in Auto/Manual	2		
		DI1, applies in Auto only	3		
		DI2, applies in Auto only	4		

Note

While the external discrete switching signal is active, it is not possible to return to the auto mode manually.

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Example:



One application of external auto/manual switching is cascade control. Two controllers may be connected as shown in the figure. Controller 2 is the primary controller. Controller 1 handles switching and manual control through DO1 connected to the primary controller. The primary controller's output signal is prevented from going to an end point during manual control. The type of connection shown in the figure, keeps the controllers electrically isolated.

5.5 Output Signal Assignments

The DPR900 is capable of providing two analog and six discrete outputs for communication with external devices.

5.5.1 Analog Output (AO)

The controller can send output signals to the process over either of two analog output channels (AO1 and AO2) or over one or two discrete output channels (DO5 and DO6). The output signal from the PID algorithm may be one of three types selected by using FCODE 102.

Function	FCODE	Option	No	Default Value	Selected Value
Con-troller Signal	102	Analog output Positioning motor output (DO5 and DO6) Pulse-width modulated output (DO5)	1 2 3	1	

5.5.1.1 Analog Output (Option 1)

Choose analog outputs to provide signals to the process in the 4–20 mA or 0–20 mA range. Set the signal range using FCODE 123 (AO1) and 124 (AO2).

Function	FCODE	Option	No	Default Value	Selected Value
Signal Range AO1	123	4–20mA output signal 0–20mA output signal	1 2	1	
AO2	124	4–20mA output signal 0–20mA output signal	1 2	2	

To configure output signal limits, use FCODE 105.

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Function	FCODE	Option	No	Default Value	Selected Value
Output Signal Limits AO1	105	Disabled	0	0	
		Signal limits apply in Auto /Manual	1		
		Signal limits apply in Auto only	2		

To set the individual High and Low signal limits, use FCODE 7 and 8.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Output Signal Limits AO1	Lower limit	7	0 – 100	%	1%	0	
	Upper limit	8	0 – 100	%	1%	100	

Use FCODE 115, Option 5 or 6 to automatically select either the Min or Max signal between the controller's output signal and an externally supplied output signal.

Function	FCODE	Option	No	Default Value	Selected Value
Output Signal Processing	115	Disabled	0	0	
		. . . . .	1		
		Min selection enabled	5		
		Max selection enabled	6		

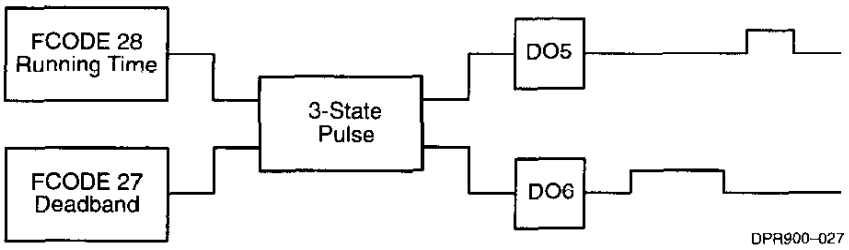
To select an internal signal for output to a recorder or other device via AO2, use FCODE 114.

Function	FCODE	Option	No	Default Value	Selected Value
Output Selector AO2	114	Disabled	0	0	
		Process value	1		
		Active setpoint	2		
		Local setpoint	3		
		Calculated Remote setpoint	4		
		Internal output signal	5		

5.5.1.2 Positioning Motor Output (Option 2)

This option provides PID positioning signals via discrete outputs DO5 and DO6. Use this option when connecting to electric actuators (reversible control motors). The discrete outputs are normally connected to intermediate relays or contactors (if the load is inductive, a diode must be wired across the output). The discrete outputs supply raise/lower pulses to a positioning motor. The minimum resolution is one sampling period, 0.2 seconds.

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When the user sets FCODE 102 to Option 2, the controller sends raise/lower pulses out on signal lines DO5 (increasing output) and DO6 (decreasing output). If both discrete signals are active at the same time, the controller takes no action.

Note

When using the positioning motor output option, the user must ensure that the controller's PID parameters are configured to have integral (reset) action. This option cannot function in **P only** or **PD only** controller applications.

Use FCODE 27 to specify a positioning motor error signal deadband. Any required controller adjustment shorter than the chosen deadband does not result in controller regulation. In addition, the controller incorporates a block that prevents the control motor from changing direction more often than every other second.

Use FCODE 28 to set the controller running time. This is defined as the actual **ON** time required to stroke the valve/motor from 0 to 100% or 100 to 0%.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Options For Output Positioning Motor	Raise/Lower Deadband	27	0–9.9	%	0.1%	0.5	
	Running time for 0–100% change	28	1–999	s	1s	60s	

Several available selectable controller functions require the use of an actuator feedback potentiometer for the function to be fully utilized. Those functions include Autotuning, Feedforward, Restart value, output signal limiting and forced control. The actuator signal will be shown on the small bargraph on the right side of the faceplate. If no position feedback is available to the controller, the bargraph displays 0%. Use FCODE 133 to enable the actuator feedback option.

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Function	FCODE	Option	No.	Default Value	Selected Value
Actuator Feedback Potentiometer	133	Not connected Connected to AI4 Connected to AI3	1 2 3	1	

Note

Fisher Controls recommends the use of actuator feedback whenever possible. If no actuator feedback potentiometer is used, the user must short AO1 on the connection terminal block. This will prevent an open-circuit alarm (Err 4) from being generated.

If required, the controller can be configured to supply the feedback potentiometer with current from AO1 (when not used for other purposes). Use FCODE 29 to select the current supplied to the potentiometer.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Potentiometer Feedback	Current output	29	0–20	mA	0.1mA	5mA	

The potentiometer signal received at AI4 can be amplified by using FCODE 30 The selected current multiplied by the potentiometer resistance must result in a voltage drop of 5V (*i.e.* the feedback signal from the potentiometer must be 0–5V for 1–100% actuator positioning.

Use FCODE 31 to set a zero-level offset for the potentiometer.

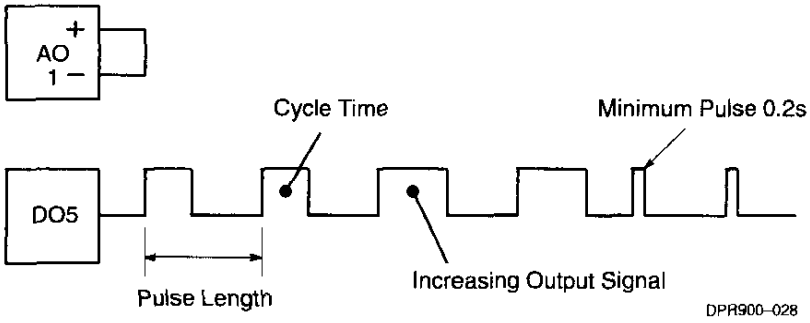
Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Actuator Feedback Signal	Gain Zero offset	30 31	0–9.99 +100	– %	0.01 1%	1 0	

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5.5.1.3 Pulse Width Modulated Output (Option 3)

This option provides for discrete output of the PID signal in a two-stage pulse-width modulated format through DO5. This output should be connected through an intermediate relay or contactor. If the load is inductive, a diode must be wired across the output. An increasing control signal forces DO5 to conduct for a longer percentage of the specified cycle time. At the maximum output signal (100%), the pulse width equals the selected pulse cycle time and results in a continuous **ON** signal. The cycle time must be selected in relation to the time constant of the process and should be shorter than five time-constants. The minimum resolution is one controller sampling period (0.2 seconds).

Use FCODE 32 to define the total cycle time.



Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Pulse-Width Output Option	Cycle time	32	1–999	s	1s	100	

Note

When pulse-width modulated output is selected, AO1 must be shorted on the connection terminal block (Pins 9 and 10).

5.5.2 Discrete Outputs

The DPR900 Controller provides for the use of 24Vdc discrete outputs to be used for external indication of switching. Select active state definition for DO1–4 using FCODE 129 thru 132. The controller defines a discrete output signal as active when conducting or open.

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Function	FCODE	Option	No	Default Value	Selected Value
DO1 (A/M)	129	Open	1	1	
Active State		Closed	2		
DO2 (R/L)	130	Open	1	1	
Active State		Closed	2		
DO3 (Alm)	131	Open	1	1	
Active State		Closed	2		
DO4 (Alm)	132	Open	1	1	
Active State		Closed	2		

5.6 Process Value (PV) Handling

Always hard wire the PV to the controller at analog input AI1. The controller can monitor, filter, calculate or linearize this value before it is employed in the controller algorithm. Analog-to-digital (A/D) conversion of the PV has an accuracy of 0.025 percent of the 0 to 20mA input.

FCODE 143 value determines how the process value appears on the digital display. Set this value to 1 for a display in physical units, and 2 thru 5 for percentage display.

Function	FCODE	Option	No	Default Value	Selected Value
Normal Display	143	Display process value	1	1	
		Display AI1, in percent	2		
		Display AI2, in percent	3		
		Display AI3, in percent	4		
		Display AI4, in percent	5		

5.6.1 Filtering Analog Inputs

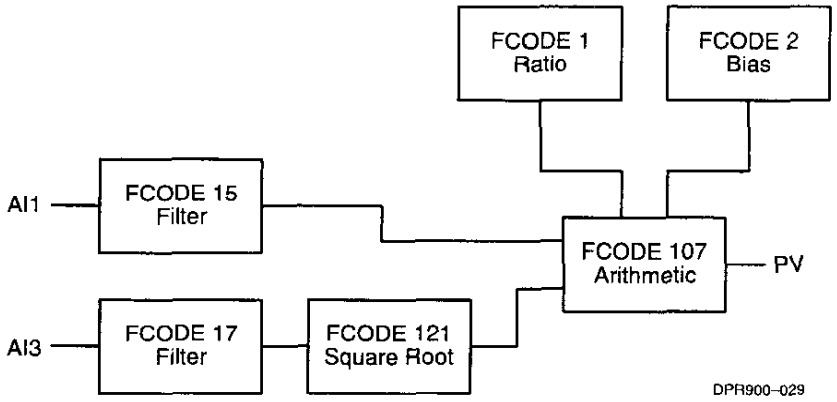
Select a first-order, low-pass filter for the individual PV inputs via FCODE 15 thru 18. Set the value of the FCODE to indicate the analog time constant in seconds, from 0 to 9999. Set the value to 0 to disable the filter.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Analog Input Filter	Time constant AI1	15	0–9999	s	1s	0	
	Time constant AI2	16	0–9999	s	1s	0	
	Time constant AI3	17	0–9999	s	1s	0	
	Time constant AI4	18	0–9999	s	1s	0	

5.6.2 Process Value Calculation

The analog input values can be conditioned to provide a required input to the controller. Square-root linearization can be applied, for example, in flow measuring with differential pressure sensors. The control provides onboard selectable square-root extraction.

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Use FCODE 107 to apply square-root linearization or use the arithmetic function for AI1.

Function	FCODE	Option	No	Default Value	Selected Value
Calculation of Process Variable	107	Disabled	0	0	
		Square root (AI1)	1		
		R = (AI1 x Ratio) + Bias	2		
		R = AI1 + (AI3 x Ratio) + Bias	3		
		R = (AI1 x Ratio) + AI3 + Bias	4		
		R = (AI1 x AI3)(Ratio) + Bias	5		

Apply square-root linearization to analog inputs AI2 thru AI4 using FCODE 120 thru 122.

Function	FCODE	Option	No	Default Value	Selected Value
AI2	120	4–20mA, 1–5V	1	1	
		0–20mA, 0–5V	2		
		4–20mA, 1–5V, square root	3		
		0–20mA, 0–5V, square root	4		
AI3	121	4–20mA, 1–5V	1	1	
		0–20mA, 0–5V	2		
		4–20mA, 1–5V, square root	3		
		0–20mA, 0–5V, square root	4		
AI4	122	4–20mA, 1–5V	1	1	
		0–20mA, 0–5V	2		
		4–20mA, 1–5V, square root	3		
		0–20mA, 0–5V, square root	4		

Use FCODE 1 and 2 to set Ratio and Bias levels.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Ratio (AI2)	Gain	1	0–9.99	–	0.01	1	
Bias (AI2)	Zero level offset	2	± Meas. Range	any	1 digit	0	

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Note

If the arithmetic function (FCODE 107) is applied to the process value AI1, the remaining arithmetic function (FCODE 108 — RSP calculation) cannot be used at the same time.



Note

If the square-root function is used on AI1 in conjunction with selecting PV monitoring, the second analog input (AI3) must have the same square root measuring range and engineering unit values. The square-root function cannot be used with a calculation.

5.6.3 Process Value Monitoring and Alarms

Two sensors can be used in applications where a sensor fault could have serious consequences. The controller can be configured to receive two transmitter input signals (AI1 and AI3) and monitor the status of these signals. The signals are scanned every 200 ms with the averaged values used as the PV for the PID algorithm. If one of the signals drifts and the deviation becomes excessive, or a sensor is disconnected, the suspect sensor is not used.

It is important to note that the controller can be configured to utilize DO3 and DO4 in one of several ways, including: process value monitoring, process value alarm, setpoint deviation alarm, and setpoint ramping indication or alarms.

5.6.3.1 Process Value Monitoring

Configure PV monitoring using FCODE 22, 23, and 116. The controller uses the average value of the two signals connected to AI1 and AI3 in the monitoring-control algorithm. The control algorithm compares the absolute difference between the previously calculated mean sensor value and the current sensor value with the maximum error signal as configured in FCODE 22. If the resulting value for AI1 is greater than the allowed deviation, an error is reported out on DO3. If the error is for the transmitter connected to AI3, the resulting alarm is sent on DO4. Note that D03 and D04 are mutually exclusive—only one or the other is displayed, not both.

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Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Process Value Monitoring	Maximum deviation Delay time	22 23	Meas. 0—999	any s	1 digit 1s	Max 10	

The controller uses the average of the signals from the previous sample as the mean value reference.

$$\text{Mean Value} = \frac{(\text{sensor 1} + \text{sensor 2})}{2}$$

If the average value exceeds the limit, the controller disables the sensor that provided the value furthest from the limit, and the controller uses the signal from the other sensor alone.

The controller continues to monitor the disabled sensor. If that sensor later provides a value inside the deviation limit, the controller enables the sensor and returns to the averaging mode. To prevent unnecessary oscillation from disabling and enabling a sensor that is sending values near the limit, every sample has a hysteresis of 0.5 percent and a user-selectable time delay (FCODE 23).

If only one sensor is attached, disable PV monitoring using FCODE 116.

Function	FCODE	Option	No	Default Value	Selected Value
PV Monitoring, PV/RSP Value Selection	116	Disabled	0	0	
		PV Monitoring, DO3, DO4	1		
		PV Monitoring, not using DO	2		
		Min selection AI1/AI3 PV	3		
		Max selection AI1/AI3 PV	4		
		Min selection, AI2/AI3 SP	5		
		Max selection, AI2/AI3 SP	6		



Note

If a temperature transmitter is connected to AI3 (PV monitoring) as a second transmitter option (redundant sensors/transmitters), AI1 must have the same measuring range and engineering unit values.

Discrete outputs DO3 and DO4 indicate the status of the averaged analog inputs when the user selects FCODE 116, option 2. When the AI1 sensor is disabled, the controller activates DO3; when the AI3 sensor is disabled, the controller activates DO4.

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Note

DO3 and DO4 outputs can indicate only one of the following: sensor status, PV alarms, setpoint deviation alarm limits, or setpoint ramping.



Note

The use of the PV monitoring function precludes the use of the Min/Max selection function for both the PV and remote setpoint.

5.6.3.2 Process Value Alarms

There are five alarm options and two possible limits for process value alarms. Each alarm option uses the same limits. Use FCODE 103 to set type of alarm.

Function	FCODE	Option	No	Default Value	Selected Value
Process Value Alarm	103	Disabled	0	0	
		High/low alarm	1		
		High alarm only	2		
		Low alarm only	3		
		High/High-High alarm	4		
		Low/Low-Low alarm	5		



Note

Process value alarms, setpoint deviation alarms, sensor status, and setpoint ramping all use DO3 and DO4, and are mutually exclusive. If more than one of these functions is set at a time the display indicates **ERROR** during configuration.

The following table shows which outputs carry the selected alarms. Only one of the indications can be on at any given time.

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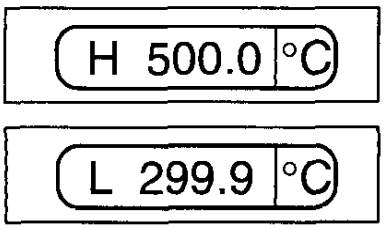
Discrete Output	FCODE 103 Options				
	1	2	3	4	5
DO3	L	–	L	H*	LL*
DO4	H	H	–	HH*	L*

* D03 and D04 are mutually exclusive. Only one alarm at a time is indicated. For example, when D04 goes on D03 goes off.

Set the low alarm limit using FCODE 3 and the high alarm limit via FCODE 4.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Process Value Alarm Limits	Limit 1 Limit 2	3 4	Meas. Range	any any	1 digit 1 digit	Min Max	

The digital display indicates a lower limit alarm for the LOW, HIGH/LOW, or LOW/LOW FCODE 103 options by an **L** that flashes to the left of the displayed value.



DPR900-030

For FCODE 103 HIGH, LOW, and HIGH/LOW options, the display indicates a higher limit alarm by a flashing **H** or **L** to the left of the displayed value.

When the user configures HIGH/HIGH or LOW/LOW alarms the display indicates the second high alarm or the second low alarm by an **H** or **L** that flashes faster (2X) than the normal alarm display.

Recall that the PV bargraph shows the configured alarm limits by brighter bars. However, it is possible to set limits outside the area shown by the bargraph that will set off an alarm if they are exceeded.

The alarms may be sent out via discrete output DO3, for lower alarms, and DO4 for higher alarms if DO3 and DO4 are not configured to carry process value alarms, setpoint deviation alarms, setpoint ramping, or is configured to monitor sensor status. The alarms have a hysteresis of 1 percent of the configured range.



Note

Both high and low alarm limits may be set, but only one at a time is active if FCODE 103 is set to Option 2 or 3. When FCODE 103 Options 4 and 5 are configured, the two alarm limits become high/high or low/low limits.

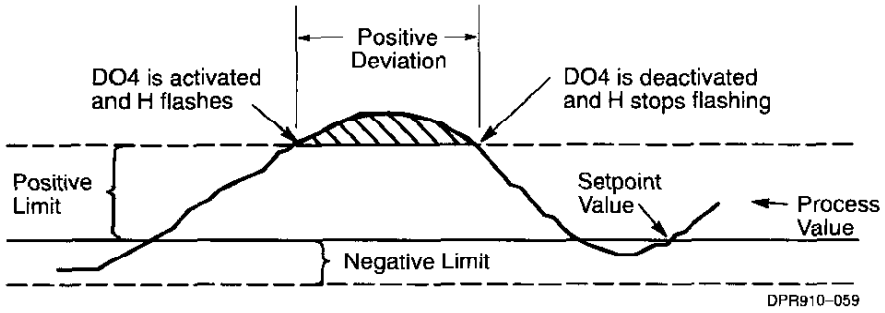
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5.6.3.3 Deviation Alarm Limits

The DPR900 is capable of externally indicating setpoint deviation alarms, provided that absolute alarms have not been selected. Configure setpoint deviation alarms using FCODE 104. The controller measures deviation and activates deviation alarms for the process value relative to the current setpoint value as shown in the figure.

These alarms have a hysteresis of 1 percent of the configured range.

Function	FCODE	Option	No	Default Value	Selected Value
Deviation Alarm Selection	104	Disabled Enabled	0 1	0	



Note

PV alarms, setpoint deviation alarms, sensor status, and setpoint ramping all use DO3 and DO4 and are mutually exclusive. If more than one of these functions is set at a time, the display flashes **ERROR**.

Use FCODE 5 to set the setpoint negative deviation limit in the range 0 to the lower measuring range limit.

Use FCODE 6 to set the setpoint positive deviation limit in the range 0 to the upper measuring range limit.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Deviation Alarm Limits	Negative limit Positive limit	5 6	0–(–M) 0–(+M)	any any	1 digit 1 digit	–Max +Max	

Brightly lit bars on the process value bargraph indicate the setpoint deviation alarm limits. The deviation alarm limits interact with the setpoint value and change automatically if the setpoint value is altered.

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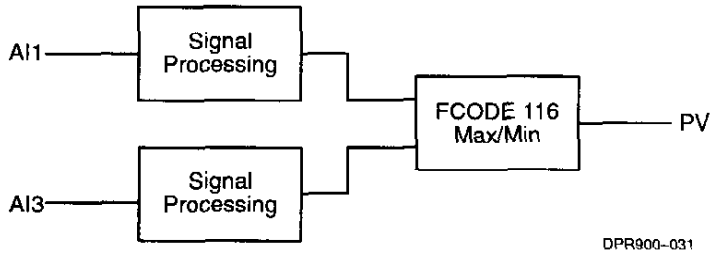
For FCODE 103 HIGH, LOW, and HIGH/HIGH options, the display indicates a higher limit alarm by a flashing **H** or **L** to the left of the displayed value.

A flashing **H** indicates a positive deviation alarm and a flashing **L** indicates a negative alarm on the digital display.

The deviation alarms may be sent out via discrete output DO3, for the negative alarm and DO4, for the positive alarm, if DO3 and DO4 are not configured to carry PV monitoring status, PV alarm, or setpoint-ramping information.

5.6.3.4 Min/Max Analog Input Selector

As an alternative to PV monitoring, the minimum or maximum signal can be selected between analog inputs AI1 and AI3 using FCODE 116, Option 3 or 4.



The selected signal is used by the controller as the PV. The Min/Max comparison is made **after** performing any signal processing functions that have been enabled.

The Min/Max function uses only analog inputs AI1 and AI3 and cannot be used if the PV monitoring function is in use.

Function	FCODE	Option	No	Default Value	Selected Value
PV Monitoring, PV/RSP Value Selection	116	Disabled	0	0	
		. . . . .			
		Min selection AI1/AI3 PV	3		
		Max selection AI1/AI3 PV	4		
		. . . . .			

5.6.4 Signal Selector — Analog Output 2 (AO2)

The user may configure the controller to send a specific internal value out on AO2 using FCODE 114. This output is scaled exactly the same as the PV/SP bargraph displays. Option 2 of FCODE 114 refers to the actual setpoint active in the controller. Option 3 is only local setpoint, regardless of the installed configuration. Option 4 is the RSP value after Ratio (FCODE 1) and Bias (FCODE 2) have been applied, when those options are configured. Option 5 is the internal output signal.

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Function	FCODE	Option	No	Default Value	Selected Value
Output Selector AO2	114	Disabled Process value Active setpoint Local setpoint Calculated Remote setpoint Internal output signal	0 1 2 3 4 5	0	

Set FCODE 124 to select the AO2 output signal range:

Function	FCODE	Option	No	Default Value	Selected Value
Signal Range AO2	124	4–20mA output signal 0–20mA output signal	1 2	2	



Note

If AO2 has been configured using FCODE 114, Option 1–5 and there is no circuit path between pins 11 and 12, the controller self-test reports an Error (E 103). The self-test is looking for the minimum 4 mA signal based on the configuration. Either configure to FCODE 124, Option 2 or complete the output circuit between Terminal 11 and 12 with a jumper wire.



Note

The AO2 output signal represents the actual bargraph value (0–100%). If the setpoint limiting function (FCODE 106) is enabled, the output signal is rescaled to represent 0–100% of the limited output span, not necessarily the full SP or PV potential.

5.7 Setpoint Handling

This section describes the five possible sources for the controller setpoint.

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1 – Local internal setpoint (LSP)
2 – Extra internal setpoint (ESP)
3 – Remote setpoint (RSP)
4 – Computer-controlled setpoint (CSP)
5 – Externally controlled via digital signals (SPC)

The user configures the setpoint source the controller uses via FCODE 142.

Function	FCODE	Option	No	Default Value	Selected Value
Automatic Setpoint Switching	142	Disabled Switches to remote setpoint Switches to extra internal setpoint	1 2 3	1	

When option 2 is configured and the **R** button is pressed, the controller switches to the RSP value.

When option 3 is configured and the **R** button is pressed, the controller switches to the ESP value.

When external remote or local switching capability is configured, an external discrete signal connected to either DI1 or DI2 can remotely activate the switch. Remote switching is selected via FCODE 111.

Function	FCODE	Option	No	Default Value	Selected Value
External R/L Switching	111	Disabled Remote/Local switching via DI1 Remote/Local switching via DI2	0 1 2	0	

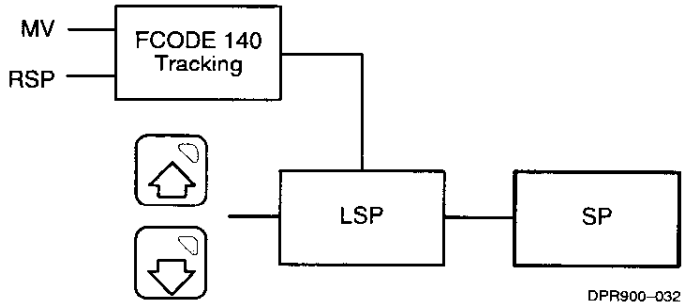
When remote switching via discrete inputs is configured and the signal line is inactive, the controller switches to LSP control. When the discrete input returns to the opposite state, the controller reverts to using the RSP value (assuming the **R** button is still lit).

5.7.1 Local Setpoint (LSP)

When no setpoint options are configured, the controller accepts a local setpoint (LSP) value, entered with the green increase/decrease buttons. Set the LSP value according to the instructions in Section 3, Establishing a Setpoint.

A local tracking option is available for selection when using LSP control. This option provides automatic setpoint tracking when the controller has been placed in the manual mode via the manual button.

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Use FCODE 140 to select the LSP tracking option.

Function	FCODE	Option	No	Default Value	Selected Value
Setpoint Tracking (LSP)	140	Disabled Tracks remote setpoint (RSP/ESP) Tracks PV in Manual	1 2 3	1	

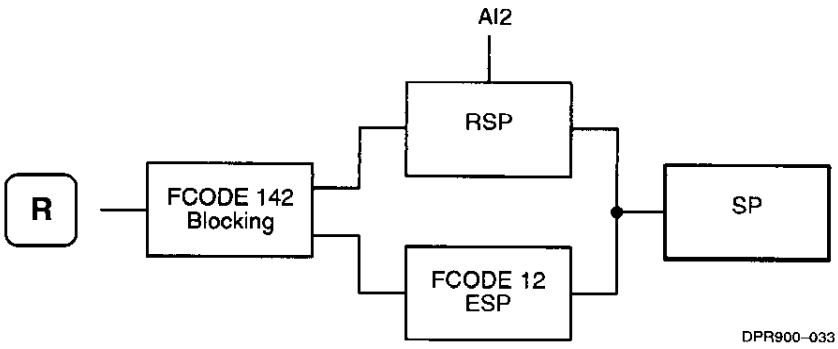


Note

The LSP tracks the PV in manual ONLY when the manual command comes from the front panel. If manual mode is forced via a discrete input, Option 3 does not take effect.

5.7.2 Extra Internal Setpoint (ESP)

The user may set a fixed extra internal setpoint value or configure the extra internal setpoint value to track a remote setpoint value. In case of computer failure, the controller continues to operate, using either the fixed extra internal setpoint value or the last RSP value received.



Use FCODE 12 to select the fixed value option.

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Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Extra Internal Setpoint	Setpoint value	12	Meas. Range	any	1 digit	Min	

Note

FCODE 12 defaults to the lower measuring range value as set in FCODE 24.

Function	FCODE	Option	No	Default Value	Selected Value
Automatic Setpoint Switching	142	Disabled Switches to remote setpoint Switches to extra internal setpoint	1 2 3	1	

The ESP is functionally a RSP. The **R** button LED remains lit when this setpoint is active, as a reminder that manual setpoint entry is not possible. If FCODE 142 is set to Option 3, switching to extra internal setpoint enabled, the extra internal setpoint is displayed in the remote mode. Simply press one of the green setpoint increase/decrease buttons to view the current digital value of ESP when active.

The local setpoint can be made to track both ESP and RSP. Switching can be done manually or using a discrete input signal. FCODE 140 is used in this case, however the ESP replaces the RSP in the description. Select the LSP tracking option using FCODE 140.

Function	FCODE	Option	No	Default Value	Selected Value
Setpoint Tracking (LSP)	140	Disabled Tracks remote setpoint (RSP/ESP) Tracks PV in Manual	1 2 3	1	

Note

The manual button overrides both the remote and extra internal setpoints, when they are active. When the controller is in the manual mode, the local setpoint can be set but is not in use.

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5.7.3 Remote Setpoint (RSP)

The DPR900 can be configured to receive an externally generated signal and use it as a remote setpoint (RSP). The controller receives the RSP value via analog input AI2 and uses this, or the extra internal setpoint, as the setpoint value when the user pressed the **R** button.

The user may condition the incoming remote signal using the following functions:

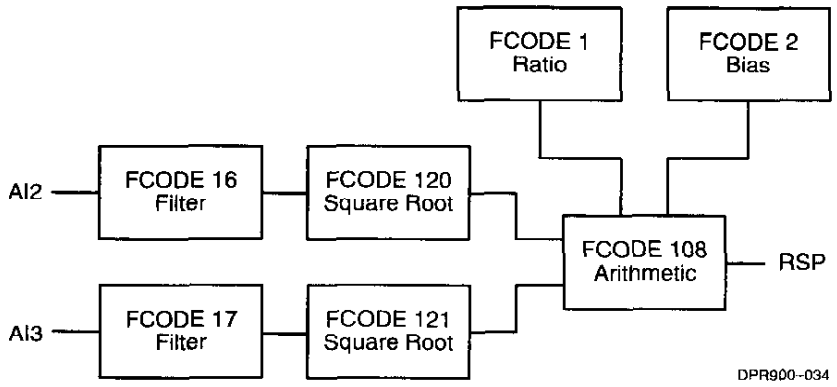
Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Ratio (AI2)	Gain	1	0–9.99	–	0.01	1	

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Bias (AI2)	Zero level offset	2	± Meas. Range	any	1 digit	0	

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Filter AI2	Time constant	16	0–9999	s	1s	0	

Function	FCODE	Option	No	Default Value	Selected Value
Signal Range (AI2)	120	4–20mA, 1–5V	1	1	
		0–20mA, 0–5V	2		
		4–20mA, 1–5V, Square root	3		
		0–20mA, 0–5V, Square root	4		

An arithmetic expression (or calculation), can be selected using FCODE 108. The result is used as the RSP input. The values of Ration and Bias are determined by FCODE 1 and 2.



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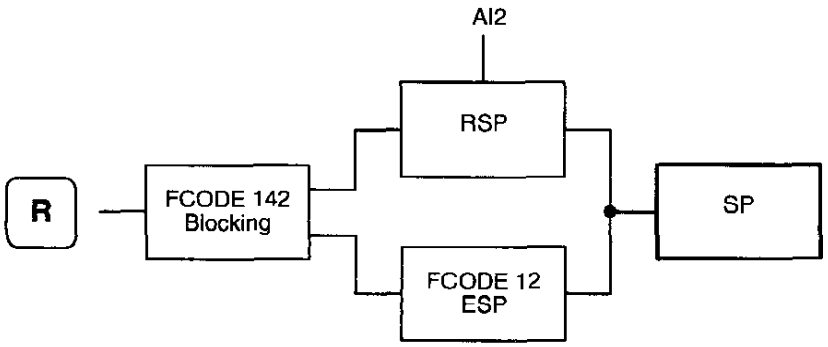
Function	FCODE	Option	No	Default Value	Selected Value
Calculation of Remote Setpoint	108	Disabled R = (AI2 x Ratio) + Bias R = AI2 + (AI3 x Ratio) + Bias R = (AI2 x Ratio) + AI3 + Bias R = (AI2 x AI3)(Ratio) + Bias	0 1 2 3 4	0	

Note

If an arithmetic function has been selected for the PV (FCODE 107 — PV calculation), this function cannot be used. Two arithmetic functions cannot be used simultaneously for both PV and RSP.

Note

AI2 can be linearized to the square root law through FCODE 120. This operation is performed before any arithmetic function is applied.



The controller receives the remote setpoint value via AI2. When the **R** button is pressed, the RSP or ESP, when configured to do so, is used as the setpoint value, depending on the setting of FCODE 142.

Function	FCODE	Option	No.	Default Value	Selected Value
Automatic Setpoint Switching	142	Disabled Switches to remote setpoint Switches to extra internal setpoint	1 2 3	1	

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If required, the digital display can be forced to show the actual RSP value rather than the default PV value. Set FCODE 143 to Option 3 to show the setpoint value received on AI2 in the digital display.

Function	FCODE	Option	No	Default Value	Selected Value
Process Value	143	Display process value	1	1	
		Display AI1, in percent	2		
		Display AI2, in percent	3		
		Display AI3, in percent	4		
		Display AI4, in percent	5		



Note

FCODE 143 Option 1 always displays the scaled process variable as modified by the setpoint limiting FCODE. Options 2 – 5 always indicate the raw input values found at the terminals after filtering an linearization.

By using a discrete input signal, the controller can be externally switched to an internally stored value:

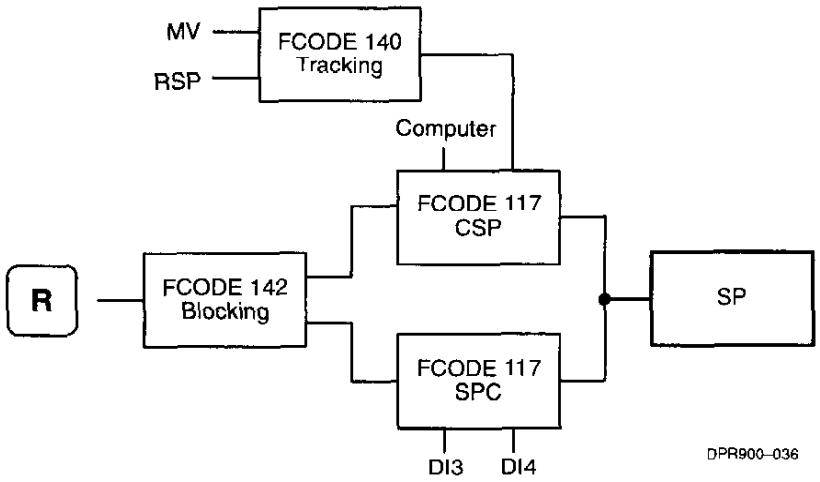
- Controller must be in the Remote mode, **R** button lit.
- If an external signal, connected to either DI1 or DI2 with FCODE 111, goes from the active to the inactive state, the controller switches to the ESP value. The **R** button remains lit as a reminder that manual setpoint manipulation is not possible.

If the input returns to the active state, the remote value is switched back in. Switching between Remote mode (LED lit) and internal mode (LED not lit) can be done by means of the **R** button. DO2 is active when the **R** button is lit.

5.7.4 Computer-Generated Setpoint (CSP)

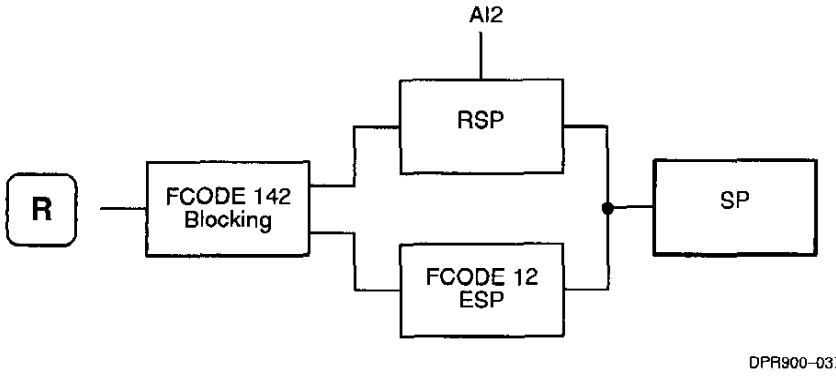
See paragraph 5.9, Computer Control, for complete details for computer controlled setpoint options.

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5.7.5 Remote/Local Setpoint Switching

Put the controller in remote mode, enabling the configured remote setpoint options, by pressing the **R** button. This enables either an external remote setpoint or the extra internal setpoint. The user chooses the setpoint to be enabled via FCODE 142.



Function	FCODE	Option	No	Default Value	Selected Value
Automatic Setpoint Switching	142	Disabled Switches to remote setpoint Switches to extra internal setpoint	1 2 3	1	

A discrete input signal connected to DI1 or DI2, and configured using FCODE 111, may switch the controller from the remote setpoint to the extra internal setpoint value in the following situation:

- The controller is in the remote mode.
- The connected and configured discrete input signal becomes active.

In this situation the **R** button remains lit to indicate that manual control is not possible. DO2 is active when the **R** button is lit as an external indicator of the controller mode.

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When Option 2 is configured and the **R** button is pressed, the controller switches to the RSP value. When Option 3 is configured and the **R** button is pressed, the controller uses the ESP value.

Function	FCODE	Option	No	Default Value	Selected Value
External R/L Switching	111	Disabled Remote/Local switching via DI1 Remote/Local switching via DI2	0 1 2	0	

When remote switching via discrete inputs is configured and the signal line is inactive, the controller switches to LSP control. When the discrete input returns to the active state, the controller reverts to using the RSP value.

Note

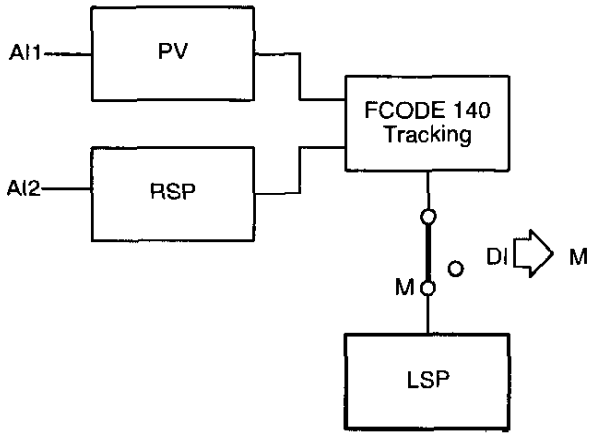
The discrete inputs are only able to switch the controller from remote to local mode. The **R** button is the only way to return the controller to remote from local mode.

5.7.6 Setpoint Tracking

The user may configure the ESP to track the RSP via FCODE 140, Option 2. Switching may be accomplished manually or automatically via an external discrete signal. The tracking function, however is enabled only with a manually entered command.

Setpoint tracking allows the user to configure the LSP to track the PV when the controller is in the Manual mode, provided the **M** button has been pushed or the controller has been switched to manual via an external computer. It does not track if the controller has been switched via a discrete input.

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Upon return to the auto mode, the LSP value will be the same as the PV signal. Use FCODE 140 to select tracking of PV in the manual mode.

Function	FCODE	Option	No	Default Value	Selected Value
Setpoint Tracking (LSP)	140	Disabled Enabled, tracks RSP/ESP Enabled, tracks PV in Manual	1 2 3	1	

Note

The local setpoint tracks the PV in manual ONLY when the manual command comes from the front panel or via computer. If manual mode is forced via a discrete input, Option 3 **does not** take effect.

5.7.7 Setpoint Limits

The user may configure operational limits to the setpoint value, to prevent the controller from using a setpoint value outside such configured limits. The limits apply to both local and remote setpoint values.

Function	FCODE	Option	No	Default Value	Selected Value
Setpoint Limits	106	Disabled Enabled	0 1	0	

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Setpoint Limits	Low limit High limit	9 10	Meas. Range	any any	1 digit 1 digit	Min Max	

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Note

Changing the SP limits also changes the 0–100 percent scales used by the SP and PV bargraphs to that of the setpoint limits: 0 percent = lower limit and 100 percent = upper limit.

Note

When setpoint limiting is used, and the user elects to retransmit the SP signal out on AO2, the transmitted signal will be the rescaled 0–100% signal.

5.7.8 Setpoint Ramp and Hold

Setpoint ramping may be required for slow starting systems, such as a steam system that requires warming up. The user selects initial setpoint ramping options via FCODE 109. The controller may ramp either the remote or the local setpoint value and can be configured to indicate Ramp Status with DO3 and DO4.

Function	FCODE	Option	No	Default Value	Selected Value
Setpoint Ramping	109	Disabled Enabled, output on DO3–DO4 Enabled, no indicated output	0 1 2	0	

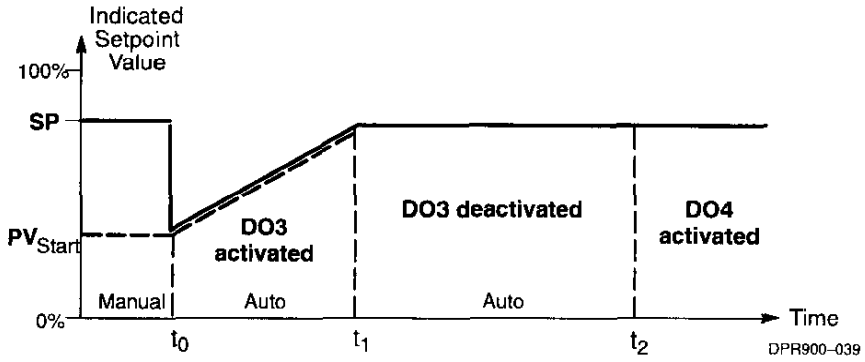
When ramping is configured, the ramp speed and ramping time period must be set via FCODE 13 and 14.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Setpoint Ramping	Ramping rate Ramping timer	13 14	Meas. 0–999	Min. Min.	1 digit 1 digit	Max 1 Min	

The controller determines the ramp direction (up or down) depending on whether the setpoint value lies above or below the current process value. The ramp starts when the controller is switched from manual to auto. This switching can be manual or done via the control signal at DI1 or DI2.

The following shows an example of the controller ramping function.

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The heating time is 7 hours (420 minutes) and:

- The measuring range is –50 to 600 degrees C.
- The current process value is 90 degrees C.
- The final setpoint value is 300 degrees C.

The process must heat to 210 degrees C in 420 minutes, giving a ramp of 0.50 degrees C/minute. The figure shows the heating time between t_0 and t_1.

At t_0 the setpoint value equals the process value. The controller ramping algorithm moves the setpoint value at a constant speed to the preset value during t_0 to t_1. The PID parameters are not influenced by the ramp speed, and the actual value follows the changing setpoint value throughout the ramping.

The ramp starts when the controller switches from manual to auto, t_0. When the controller is switched to auto, the setpoint ramping starts at the present process value, moving at the preset speed towards the final setpoint.

DO3 and DO4 indicate the ramp status when FCODE 109 is set to 1. DO3 is active during ramping, from t_0 to t_1. DO4 is active at t_2, indicating that the configured time period is elapsed.

While the ramp is in control, the digital display shows the instantaneous value of either the process value or the setpoint, depending on the configuration. The final setpoint value displays as a brighter bar on the bargraph and the PV and SP bargraphs also show the instantaneous values.

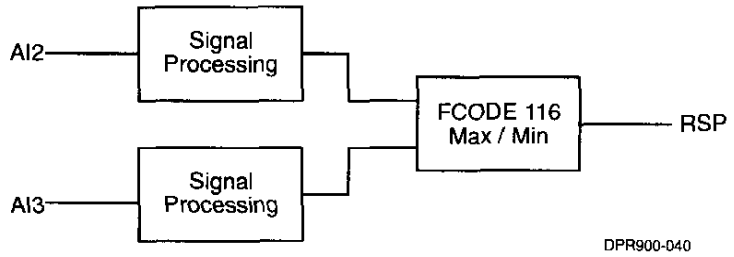
To alter the setpoint value while the ramp is active, switch the controller to manual mode. Insert the new setpoint value and restart the ramp with the new parameters.

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The setpoint value limit FCODE 106 can be used to include the area of most interest in the bargraph range. If further indications are required as certain levels are passed, the process value alarms can be used, provided the digital outputs have not been assigned previously.

5.7.9 Min/Max RSP Selection

The Min/Max selector function is used to select one of the two RSP values connected on AI2 and AI3. The selected signal will then be used as the RSP signal. The Min/Max comparison is made after any signal processing that may have been applied.



The RSP Min/Max function (FCODE 108) can only be used for the PV or the RSP by excluding the use of the PV monitoring function.

Function	FCODE	Option	No	Default Value	Selected Value
PV Monitoring, PV/RSP Value Selection	116	Disabled	0	0	
		PV monitoring DO3, DO4	1		
		PV monitoring not using DO	2		
		Min selection AI1/AI3 PV	3		
		Max selection AI1/AI3 PV	4		
		Min selection AI2/AI3 RSP	5		
		Max selection AI2/AI3 RSP	6		



Note

The Min/Max functions for PV and RSP and the PV monitoring functions are mutually exclusive and cannot be combined.

5.8 Analog Output Signal Handling

This section describes the controller functions the user may select to apply to the analog output signals AO1 and AO2.

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Note

Manual control of output signals is only possible when the **M** button and the manual mode increase /decrease buttons are active (yellow LED's lit). When only the **M** button is lit, the controller has been forced to the manual mode externally, i.e. with discrete switching signals or through the RS-485 communications link.

5.8.1 Manual Output Signal Control (AO1)

Press the **M** button to select the manual mode. The yellow LED's in the **M** button and the increase/decrease buttons light to indicate Manual mode. The user can then change the output signal in 0.5 percent increments, using the yellow increase/decrease buttons.

To return to the Auto mode, simply press the **M** button again. The yellow LED goes out, indicating Auto mode.

5.8.2 Output Signal Limiting (AO1)

To prevent output signals on AO1 from exceeding specified limits, set upper and lower output signal limits, using FCODE 7 and 8. Select output signal limiting using Fcode 105. Output signal limiting does not apply to raise/lower outputs.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Output Signal Limits (AO1)	Lower limit Upper limit	7 8	0-100 0-100	% %	1% 1%	0 100	

Function	FCODE	Option	No	Default Value	Selected Value
Output Signal Limits AO1	105	Disabled Signal limits apply in Auto /Manual Signal limits apply in Auto only	0 1 2	0	

5.8.3 Forced Output Control (AO1)

With the exception of the raise/lower (positioning motor) outputs, the controller's output signal AO1 may be forced to a preset value by a discrete input at DI1 or DI2.

Enable or disable the forced control function via FCODE 112.

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Function	FCODE	Option	No	Default Value	Selected Value
Forced Output Control	112	Disabled	0	0	
		DI1, applies in Auto/Manual	1		
		DI2, applies in Auto/Manual	2		
		DI1, applies in Auto only	3		
		DI2, applies in Auto only	4		

Use FCODE 19 to set the required output signal AO1 for forced control in the range 0-100 percent.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Forced Control	Output Signal (AO1)	19	0-100	%.	1%	0	

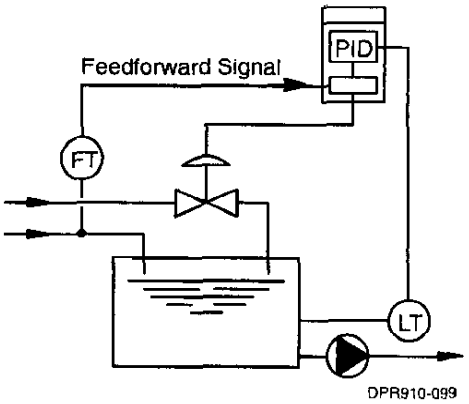


Note

If a three-state pulse output is used, actuator position feedback is required.

5.8.4 Feedforward Control (AO1)

The analog input signal at AI3 or AI4 can be amplified and added to the output signal at AO1. This function can be used to eliminate a measurable incoming disturbance before it has time to influence the process value.



The feedforward function sums the PID output signal with either AI3 or AI4. To apply the feedforward function, select the optional amplification of the analog signal at AI3 or AI4, and sum the amplified signal with AO1, using FCODE 115.

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Function	FCODE	Option	No	Default Value	Selected Value
Output Signal Processing	115	Disabled	0	0	
		Feedforward enabled, AI3	1		
		Feedforward enabled, AI4	2		
		Tracking, Output DI (AI4)	3		
		Tracking, Signal Selector (AI4)	4		
		Min selection enabled	5		
		Max selection enabled	6		

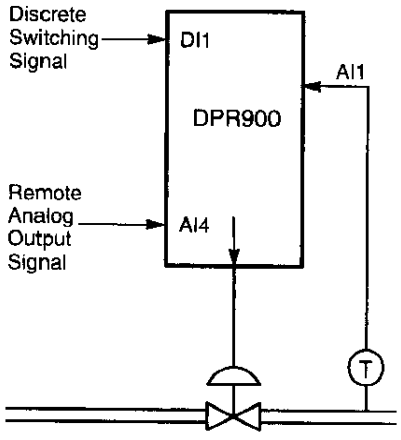
Use FCODE 11 to set the amplification in the range of 0 to ± 9.99.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Feedforward	Gain	11	0– ± 999	–	0.1	0	

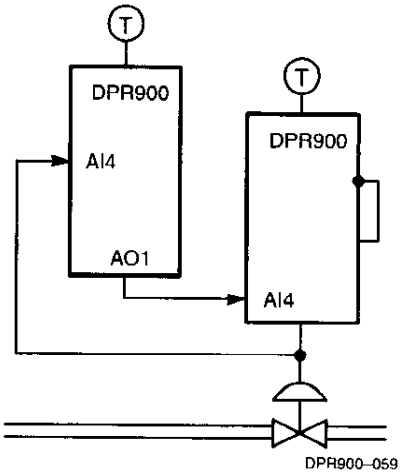
5.8.5 Output Signal Tracking

There are two types of output tracking functions selectable with the DPR900 controller using FCODE 115. While both functions deal with the output signal to the final control device, one option is **discretely selectable** through DI1 while the other option is **High/Low Signal** selectable.

Option 3: "Output Tracking"



Option 4: "Signal Selection"

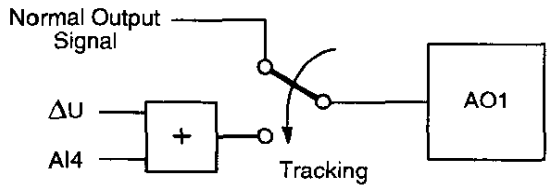


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5.8.5.1 Tracking Signal Selector (Option 3)

Output tracking option 3 is used when the controller is configured to be used as a remotely controlled **output tracking** controller. This occurs when a second signal received at AI4 is to be passed directly through the DPR900 and used as the output signal to the final control device. As illustrated, when the discrete input signal is **active**, the controller takes the analog signal at AI4 and retransmits it directly through analog output AO1 to the control valve. With this selection active, internal output signal tracking is automatically achieved for the internal loop.

Option 3 should be selected when a remote analog signal is to be retransmitted through the controller to the final control device based on a discrete switching signal. The controller's output signal can be made to track a given analog input signal connected to AI4, providing output tracking when the discrete is active.



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When output tracking is enabled, the output signal at AO1 = (AI4 + delta U). Delta U is defined as the contribution from the controller during regulation (max ± 10%).

The user selects this function with FCODE 115, Option 3. The controller will track only when the controller is in the AUTO mode and the discrete input at DI1 is active. If the controller is in the MANUAL mode, the discrete input has no effect on the controller switching action and the controller remains manually adjustable.

The output tracking function cannot be used with positioning motor (raise/lower) pulse outputs.

Function	FCODE	Option	No	Default Value	Selected Value
Output Signal Processing	115	Disabled	0	0	
		Feedforward enabled, AI3	1		
		Feedforward enabled, AI4	2		
		Tracking, Output DI (AI4)	3		
		Tracking, Signal Selector (AI4)	4		
		Min selection enabled	5		
		Max selection enabled	6		

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5.8.5.2 Tracking Signal Selector (Option 4)

Option 4 should be selected when twp DPR900 controllers are used and the high (or low) analog output signal between the controllers is selected for use with the final control device by the second DPR controller.

Option 4 is used only in the controller supplying its output signal to another DPR controller and used with the second controllers Min/Max signal selection (FCODE 115, Option 5 or 6). Option 4 allows for automatic integral output tracking between the two controllers, utilizing the analog input signal connected at AI4. This signal must be the output signal selected for transmission to the final control device.

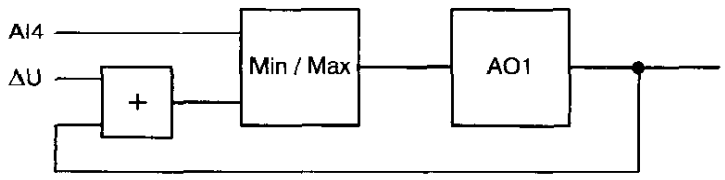
Function	FCODE	Option	No	Default Value	Selected Value
Output Signal Processing	115	Disabled	0	0	
		Feedforward enabled, AI3	1		
		Feedforward enabled, AI4	2		
		Tracking, Output DI (AI4)	3		
		Tracking, Signal Selector (AI4)	4		
		Min selection enabled	5		
		Max selection enabled	6		

This output tracking function cannot be used with positioning motor (raise/lower) outputs.

Function	FCODE	Option	No	Default Value	Selected Value
External A/M Track Switching	110	Disabled	0	0	
		Auto/Manual/Track switching via DI1	1		
		Auto/Manual/Track switching via DI2	2		

5.8.6 Output Min/Max Signal Selector (AO1)

The output min/max function lets the output signal be controller by either AI4 or the controller-calculated ΔU (delta U) value. ΔU is calculated as AO1 + ΔU. ΔU is the contribution from the controller during regulation (max ± 10%).



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The min/max selection is only active in the auto mode. Additionally, the function cannot be used in conjunction with positioning motor (raise/lower) pulse outputs.

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Function	FCODE	Option	No	Default Value	Selected Value
Output Signal Processing	115	Disabled	0	0	
		Feedforward enabled, AI3	1		
		Feedforward enabled, AI4	2		
		Tracking, Output DI (AI4)	3		
		Tracking, Signal Selector (AI4)	4		
		Min selection enabled	5		
		Max selection enabled	6		

5.8.6.1 Output Signal Display

The displayed output signal may be **reversed** (i.e. 0—100% actually displayed as 100—0%). This may be useful when the operator is familiar with an increase to close situation (reverse acting) and the displays associated with such a loop. Use FCODE 134 to select the values displayed as 0—100% or 100—0%.

Note

Reversing the output signal display affects only the display. The up and down arrow keys and all other control functions remain unchanged.

Function	FCODE	Option	No	Default Value	Selected Value
Output Signal Display	134	0—100%	1	1	
		100—0%	2		

5.8.7 External Setpoint Output (AO2)

The user may configure the controller to send the active setpoint value, whether remote, local or computer controlled, out on analog output 2 (AO2) using FCODE 114. Option 2 refers to the actual active setpoint value of the control loop. Option 3 always refers to the local setpoint value regardless of the configuration. The remote setpoint at AI2 can also be retransmitted to AO2 **after** signal processing (i.e. filtering, square-root extraction, and arithmetic handling).

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Function	FCODE	Option	No	Default Value	Selected Value
Output Selector AO2	114	Disabled	0	0	
		Process value	1		
		Active setpoint	2		
		Local setpoint	3		
		Calculated Remote setpoint	4		
		Internal output signal	5		



Note

The controller self-test generates an error (E103) if FCODE 114 is other than 0 and AO2 terminals (11 and 12) are open.

Set FCODE 124 to select the AO2 output signal range.

Function	FCODE	Option	No	Default Value	Selected Value
Signal Range AO2	124	4—20mA	1		
		0—20mA	2	2	



Note

If the controller is using a remote setpoint value and FCODE 140 is configured for Option 1 (the extra internal setpoint value does not track the remote setpoint value), then the extra internal setpoint may be sent out on AO2 by selecting FCODE 114, Option 2.



Note

The AO1 output signal represents the actual bargraph value (0—100%). If setpoint value limiting (FCODE 106) is enabled, the output signal is rescaled to represent 0—100% of the limited output or SP span, not necessarily the full SP or PV potential.

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5.8.8 Start–Up and Fatal Fault Values

Select an output signal value for the controller to use at startup and /or configure the controller to hold the current output signal level, or go to a preset signal level in case of a fatal fault during continuous self-testing.

Use FCODE 20 to set the initial output signal start-up value for AO1 in the range 0-100 percent. The value must be within the output signal limits if any are configured.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Start Value	Output Signal (AO1)	20	0-100	%	1%	0	

Use FCODE 113 to select the start value option the controller uses when it detects a fatal controller error.

Function	FCODE	Option	No	Default Value	Selected Value
Controller Fault	113	Freeze output , at current level Set output level to preset value	1 2	1	

Use FCODE 21 to set the preset failure output signal level for AO1 in the range 0-100 percent to be used in the event of a fatal controller error as selected with FCODE 113, Option 2.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Controller Fault	Output signal (AO1)	21	0-100	%	1%	0	



Note

These functions are not applicable for positioning motor (raise/lower) outputs.

5.9 Computer Control

The following covers controller communications with a supervisory system and external control of either the setpoint or the output signal. FCODE 117 determines the data communications configuration of the controller.

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Function	FCODE	Option	No	Default Value	Selected Value
Data Commu-nications	117	Disabled Not Used Output signal control Remote setpoint control COMLI RS-485 protocol	0 1 2 3 4	0	

When the user configures any of the other options (1-4) the red **C** light comes on. Press the **R** button to activate or disable the data communications function. The data communication mode is active only when the **R** button is lit. Discrete output DO2 is activated as soon as the controller is in the remote mode.

5.9.1 Remote Output Signal Control (DDC)

FCODE 117, Option 2 is remote output signal control. Two 24V discrete input signals, DI3 and DI4, provide remote output signal control provided the controller is in the Remote mode, the **R** button LED is lit. Activation of DI3 decreases and DI4 increases the output signal. If both signals are active no change occurs in the output signal. Choose options from the following FCODE's.

Use FCODE 33 to set the maximum ON time of the discrete inputs, in the range 1-999 seconds. This is further defined as the time it takes to move the output from 0 to 100% or 100 to 0%.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
DDC Running Time	Time for 0-100% change	33	1-999	s	1s	100s	

The DI1 or DI2 discrete input signal lines can switch the controller mode from remote to local (but not from local to remote). Configure this capability by setting the appropriate value with FCODE 111.

Function	FCODE	Option	No	Default Value	Selected Value
External R/L Switching	111	Disabled Remote/Local switching via DI1 Remote/Local switching via DI2	0 1 2	0	

5.9.2 Remote Setpoint Control (SPC)

FCODE 117, Option 3, selects remote setpoint (supervisory) control. Remote control of the setpoint is then possible via activation of discrete inputs DI3 and DI4. A 24 V signal on DI3 decreases the setpoint value and a 24 V signal on DI4 increase the setpoint value. If both inputs are active simultaneously, the setpoint does not change.

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FCODE 33 sets the maximum running time for the controller to change the setpoint from 0 to 100% (or 100 to 0%). Choose options from the following FCODE's:

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
DDC Running Time	Time for 0 to 100%	33	1-999	s	1s	100s	

Function	FCODE	Option	No	Default Value	Selected Value
External R/L Switching	111	Disabled Remote/Local switching via DI1 Remote/Local switching via DI2	0 1 2	0	

Function	FCODE	Option	No	Default Value	Selected Value
Setpoint Computer Control	141	Extra internal setpoint External remote setpoint	1 2	1	

Function	FCODE	Option	No	Default Value	Selected Value
Automatic Setpoint Switching	142	Disabled Switches to remote setpoint Switches to extra internal setpoint	1 2 3	1	

External switching to local mode via DI1 and DI2 is still possible (switching to remote from local is not possible). As in any data communications mode, control of the setpoint value is only available by activating the **R** button, otherwise the controller remains in local control mode.

5.9.3 COMLI

COMLI is the communication protocol that defines how information held within the controller resident registers can be sent to a computer. This is called the RS-485 communications link. A full description of the COMLI protocol and its application in communicating with the DPR900 controller are contained in a separate manual, UM6.1: DPR-COMLI, available from you Fisher Representative. When the user selects Option 4 of FCODE 117, COMLI communications, they must enter the COMLI address for the controller in FCODE 46. No two controller's on the same link can have identical addresses.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
COMLI ID	Controller's Highway Address	46	1-255	–	——	1	

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5.10 Autotuner

The Autotuner is a selectable function that enables automatic setting of the controller PID parameters based on actual process control dynamics. When Autotuning is used in conjunction with the Gain Scheduling function, Autotuning of non-linear processes is also possible. Paragraph 5.11 describes the Gain Scheduling function.

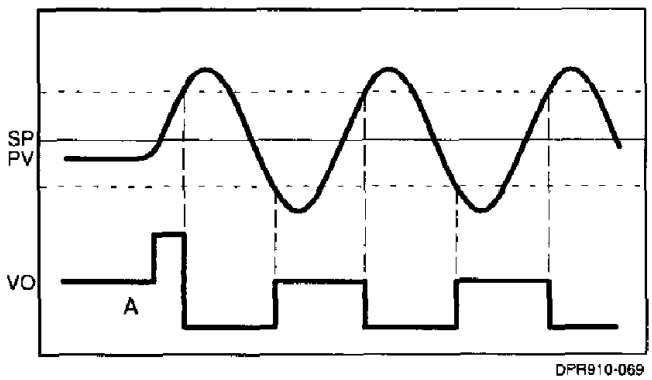
5.10.1 Principle of Operation

Autotuning uses a relay type function with hysteresis, initiating a controlled oscillation in the process. When the **TUNE** button is pressed the autotuning procedure is started by inserting a step in the output signal so that the process value begins to diverge from the setpoint. Because the relay function is controlling the process during autotuning, self-oscillation is induced, so that the output signal alternately exceeds and falls below the original value by a defined amplitude. The Autotuner itself adjusts this amplitude so that the process value will not be greater than a level which is necessary for the process response to be isolated from process and measuring noise (the oscillation amplitude is roughly comparable to the process noise level).

By measuring the process response the Autotuner can identify the process dynamic characteristics during autotuning. When the parameters are considered reliable, the autotuning process stops. New PID parameters are calculated and stored in the controller.

The autotuning function has certain similarities to the manual Ziegler Nichols method. On the other hand, the Autotuner has control over the process oscillation amplitude which results in considerably less disturbance to the process.

Tuning Process Example:



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Autotuning begins at **A.** The process noise level is evaluated and the first output signal is transmitted. When three acceptable (equal) peaks have been obtained, autotuning is complete. Autotuning can stop at any time, but no new PID parameters will be installed.

The Autotuner incorporates a form of intelligence. In situations where the D element should not be used, for example in level control, the Autotuner automatically identifies the process and eliminates the D element.

If the process has a long deadtime (at lease twice as long as the dominant time constant in the process), the normal configuration rules for PID controllers do not apply. The DPR900 has a special set-up method for dead time processes. The method involves a PI controller with relatively low gain and a short integration time.

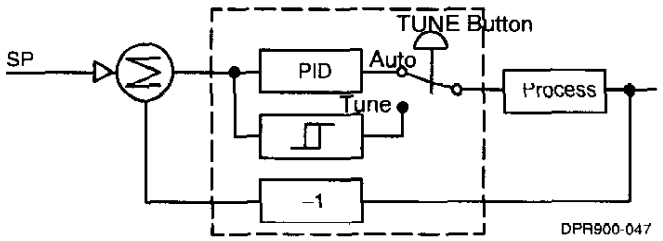
Note

It is never necessary to define the length of deadtime.

Process	FCODE 135
Most processes	Disabled
Deadtime >2 times the dominant time constant. This provides a PI controller with low gain and a relatively short I-time. This design method is specially developed and must not be used with integrating processes.	Enabled
Noisy processes which cannot be filtered.	Enabled, break on alarm

5.10.2 Initiating the Autotuner Function

Press the **TUNE** button in either the auto or manual mode. The button flashes to indicate that autotuning is in progress.



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To halt the process before tuning is complete, do one of the following:

- Press the **M** button. This switches the controller directly to the manual mode, halting the process.
- Press the **TUNE** button. The controller returns to the original state before tuning starts (auto, manual or remote). The values of the PID tuning parameters also return to their original state.



Note

Before tuning begins, the process must be stabilized near the desired working point. Autotuning should not be used for processes with long deadtimes, i.e. where process deadtimes exceed 2X the dominant time constant of the loop.

The autotuner has a time-out limit of 32 hours. If tuning ceases because this limit is exceeded, the controller returns to its original state, and displays error code 108. Fault conditions that may arise during autotuning are listed in Appendix A, error code 105 to 108.



Note

If the Autotuner function is interrupted before completion, no PID parameters are updated.

The controller returns to automatic operation when tuning is complete.

The Autotuning function may be used with any available output option, however, with positioning motor (raise/lower) output, an actuator for positional feedback must be used or the autotuning function will not work. In addition, autotuning can only be initiated by local control and the ramping function must not be in effect.

5.10.3 Autotuner Operation

The user may configure the autotuner function to be active using FCODE 135. If Option 3 is selected, the Autotuner function will automatically cease if any controller alarm becomes active during the autotuning cycle.

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Use FCODE 135 to select status of the Autotuner function.

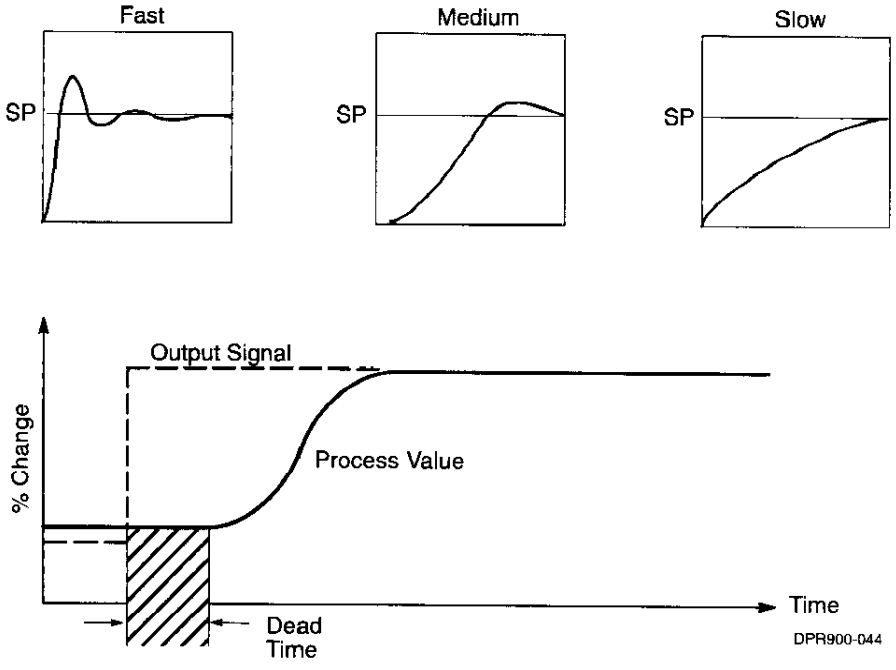
Function	FCODE	Option	No	Default Value	Selected Value
Autotuner	135	Disabled Enabled Enabled, break on alarm	0 1 2	0	

Use FCODE 136 to select maximum initial output signal change. By selecting a maximum output signal level during tuning, the user can limit the output swing generated by the controller. The user is reminded that the controller will only use the minimum swing required to solve for PID. For a majority of applications, the default value (10%) is adequate.

Function	FCODE	Option	No	Default Value	Selected Value
Max Output Signal Change	136	10 Percent 5 Percent 3 Percent	1 2 3	1	

Set a low output signal change percentage to use the autotuner for processes with moderate deadtimes.

Use FCODE 137 to select controller dynamics. After one tuning it can be used to change the dynamics without any retuning. Essentially this FCODE controls the damping factor of the function. The figure shows the dynamics of the four options.



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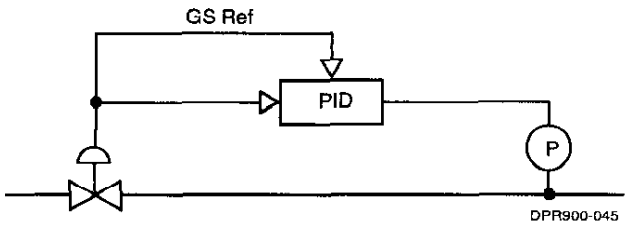
Function	FCODE	Option	No	Default Value	Selected Value
Autotune Dynamics	137	Medium, critical damping Fast, slightly underdamped Slow, slightly overdamped Deadtime	1 2 3 4	1	

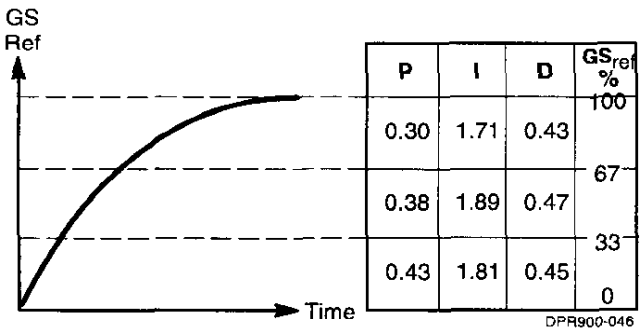
If the same controller is utilized in different processes, it is important to reset the configuration as well as the Autotuner memory. This is important in switching between fast and slow loops. Use FCODE 144 to reset the autotuner if required, for example if the controller is moved to another loop.

Function	FCODE	Option	No	Default Value	Selected Value
Resetting	144	Reset disabled Reset Autotuner Reset Gain Scheduling	0 1 2	0	

5.11 Gain Scheduling

Gain scheduling is used to improve the control of non-linear processes where the non-linearity is fixed. Processes that are not time-dependent are an example of this type of process. The figure shows an example of how gain scheduling uses a different set of PID parameters to control the process, based on the range of the selected reference signal.





Gain scheduling Principle of Operation

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5.11.1 Principle of Operation

Gain scheduling means that the controller switches between three different sets of PID parameters, depending on the value of a chosen reference signal (GSref). The reference signal may be an internal signal (the output signal or the process value) or an external signal received on either AI3 or AI4.

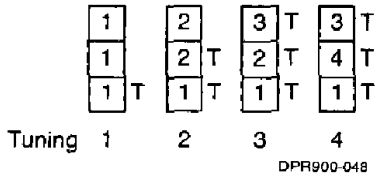
The controller stores as many as three different sets of PID parameters to use in three user-defined ranges (percent of reference signal). When the reference signal passes from one range to the next, the controller uses the set of parameters for that range.

The user may manually select and store the parameters using FCODE 36 thru 44, or use the autotuner to derive and store them.

Note

Ensure the process and reference signal levels are in the appropriate range when using the Autotuner to derive the parameters. Either drive the process manually to the required level or initiate autotuning when the process is at the appropriate level.

When the Autotuner is used to automatically set the gain scheduling table, the gain schedule reference (GSref) signal must be within the preset limits (FCODE 34 and 35) before tuning procedures begin.



The tuning sequence shows how the parameter table is completed. Note that the reference signal level changes between each tuning.

The labels 1 thru 4 at the bottom of the diagram, indicate the first thru fourth tuning cycles. Each box represents all three PID parameters. The T beside the boxes shows that tuning is complete for that range.

Note

The values derived from each tuning are placed in the ranges that have not yet been tuned. The percentage ranges given in the example are the default ranges. In the range 0-10 seconds, the increments are 0.1 seconds.

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- The first tuning, for the case where the reference signal is in the lowest range, 0-33 percent, is complete and the derived PID values are stored in all levels (FCODE 36,37,38), (39,40,41) and (42,43,44).

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
PID Parameter 1	Gain Ti (Reset) Td (Rate)	36 37 38	0.01-99.99 OFF-9999 0-9999	– s s	0.01 1s 1s	0.01 1000s 0	

- The tuning for the second range (33-67 percent) is complete after values have been derived at reference signal level 2. The derived values for this range are also stored as the values for the third level (FCODE 39,40,41) and (42,43,44).

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
PID Parameter 2	Gain Ti (Reset) Td (Rate)	39 40 41	0.01-99.99 OFF-9999 0-9999	– s s	0.01 1s 1s	0.01 1000s 0	

- The reference signal has increased to range 3 (67-100 percent) and tuning is complete. The values for this range are stored in the third level (FCODE 42,43,44).

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
PID Parameter 3	Gain Ti (Reset) Td (Rate)	42 43 44	0.01-99.99 OFF-9999 0-9999	– s s	0.01 1s 1s	0.01 1000s 0	

- A further tuning has been done when the reference signal returned to range 2. The system remembers that the tuning has been performed at all levels, so that only values at level 2 are changed.

If gain scheduling is deactivated during operation, the controller will fill the parameter with the active set of PID terms at all levels.

5.11.2 Selecting Gain Scheduling

Use FCODE 34 and 35 to set the breakpoints between the three ranges, if the default breakpoints are not appropriate.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Gain Scheduling Breakpoints	Lower Upper	34 35	0-100 0-100	% %	1% 1%	33% 67%	

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Note

When using the Gain Scheduling function, three reference ranges must be specified, even though two ranges may have the same values.

FCODE 36 thru 44 contain the PID values for the three ranges, with three values for each range.

GSref may be one of four different signals:

1. Controller's process value after signal processing.
2. Controller's actual output signal (after limiting, when selected) or the actuator feedback signal if a three-state pulse output is used.
3. Process signal or external variable received on AI3.
4. Process signal or external variable received on AI4.

Use FCODE 138 to indicate signal that will act as reference for gain scheduling.

Function	FCODE	Option	No	Default Value	Selected Value
Gain Scheduling Reference (GSref)	138	Disabled Output signal Process value Signal is AI3 Signal is AI4	0 1 2 3 4	0	

Use FCODE 144 to reset the gain scheduling. The value of this FCODE automatically returns to 0 after either resetting the gain scheduling of the autotuner.

Function	FCODE	Option	No	Default Value	Selected Value
Resetting	144	Reset disabled Reset Autotuner Reset Gain Scheduling	0 1 2	0	

After a gain scheduling reset, the controller once again records which levels have been tuned, and automatically inserts values into untuned levels. When the gain scheduling is reset, the PID values in the parameters are not erased, and the parameters remain in the tables. At the next tuning, the values obtained are used in the table.

5.11.3 Modifying Gain Scheduling Dynamics

It is possible to have different dynamics at different levels (See Parameters table). To accomplish this, change FCODE 137 before doing an Autotune function, following these steps:

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- Ensure the reference signal is at the required level.
- Use FCODE 137 to select the required controller dynamic response.
- Start tuning.

Changes to the dynamics can also be done via FCODE 137, if the current level was used for the most recent tuning.

5.12 Manual PID Parameter Setting

Note

The controller is delivered in a **de-tuned** state. No attempt should be made to use the controller without first installing the process parameters.

If the user does not have the autotuner derive the process parameters, they must be inserted manually. For normal operation enter three parameters using FCODE 36 thru 38. Enter three sets of parameters using FCODE 36 thru 44 when configuring the controller to use gain scheduling. Each PID parameter set has three parts:

Use FCODE 36, 39 and 42 to set a gain factor in the range 0.01-99.99.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
PID 1	Gain	36	0.01-99.99	–	0.01	0.01	
PID 2	Gain	39	0.01-99.99	–	0.01	0.01	
PID 3	Gain	42	0.01-99.99	–	0.01	0.01	

Use FCODE 37, 40 and 43 to set the integration time, in the range 0.1 to 9999 seconds per repeat or disable by setting to OFF.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
PID 1	Ti (Reset)	37	OFF-9999	s	1s	1000s	
PID 2	Ti (Reset)	40	OFF-9999	s	1s	1000s	
PID 3	Ti (Reset)	43	OFF-9999	s	1s	1000s	

Use FCODE 38, 41 and 44 to set the derivative time, in the range 0 to 9999 seconds per repeat. The resolution is 0.1s for intervals of 0-10 seconds and 1s for intervals between 10 and 999 seconds.

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Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
PID 1	Td (Rate)	38	0-9999	s	1s	0	
PID 2	Td (Rate)	41	0-9999	s	1s	0	
PID 3	Td (Rate)	44	0-9999	s	1s	0	

Select the type of control, where:

Ti = integration time (Reset)
Td = derivative time (Rate)
[OFF] = digital display readout

The options are:

- P (proportional only regulation); set Ti(Reset) to [OFF] and Td(Rate) to 0.0.
- PD (proportional/derivative regulation); set Ti(Reset) to [OFF].
- PI (proportional/integral regulation); set Td(Rate) to 0.0.
- PID (proportional/integral/derivative regulation); set all parameters.



Note

When gain scheduling is not configured, the PID parameters in FCODE 39 thru 44 are given the equivalent values set manually in FCODE 36 thru 38. If gain scheduling is configured, all parameters must be set, either manually or by autotuning.

The [OFF] selection for Ti (Reset) appears on the digital display after 9999. Since the range values roll over at the high and low ends, it may be easier to go backwards to reach [OFF], by using the decrease button to drop below 0000. When [OFF] is selected, the controller actually stores a value of OFF.

5.13 Direct or Reversed Controller Action

The controller is capable of providing both positive and negative gain in the process control loop. Positive gain means that an increase in output control signal results in an increase in the process value.

Use FCODE 101 to select the process gain type.

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Function	FCODE	Option	No	Default Value	Selected Value
Process Type	101	Reverse (negative gain) Direct (positive gain)	1 2	1	

	Increase Open	Increase Close
Reverse Process	Option 1 Reverse	Option 2 Direct
Direct Process	Option 2 Direct	Option 1 Reverse

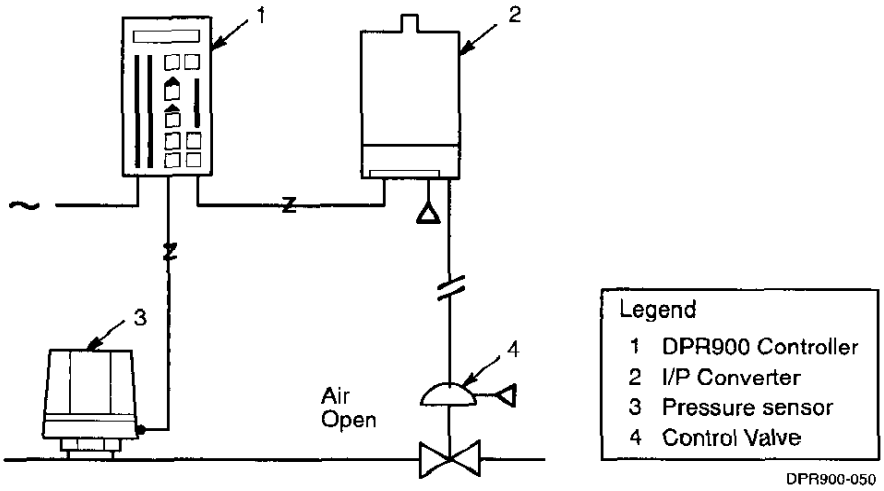
DPR900-049

Use the reverse mode for processes with positive gain; the controller supplies an output signal with negative gain. In this case, as the process value exceeds the setpoint, the output signal decreases.

Use the direct mode for processes with negative gain; the controller supplies an output signal with positive gain. As the process value exceeds the setpoint, the output signal increases.

The effect of this FCODE on raise/lower output control is:

- Direct — as the process value exceeds the setpoint, DO5 conducts, as the process value falls below the setpoint, DO6 conducts.
- Reversed — as the process value exceeds the setpoint, DO6 conducts, as the process value falls below the setpoint, DO5 conducts.

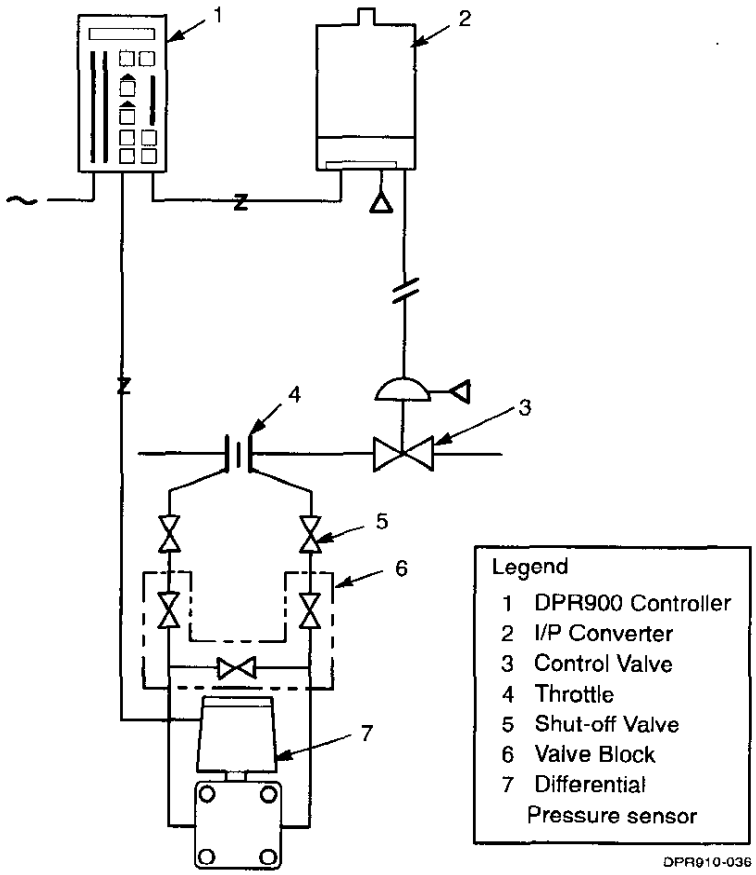


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Example 1 — Pressure Reduction

The requirement in the system shown in the following figure is for the normally open valve to close if the valve control air pressure fails. The valve also closes if pressure in the line increases. In either case an increasing process value must yield a decreasing output from the controller. The controller must be configured for reversing output.

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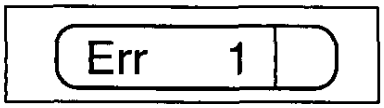
Example 2 — Flow Regulation

In the following figure the process requirement is for the normally closed valve to open if the control pressure fails. The valve also closes if the flow rate increased. In either case an increasing process value yields an increasing output signal from the controller. The controller must be configured for direct output.

5.14 Alarm and Fault Indications

The controller can monitor the process value or the deviation between the process and setpoint values and internal functions.

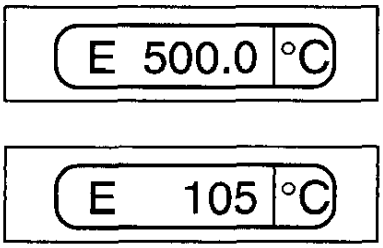
5.14.1 Fault Indication



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There are two types of fault indications, based on the severity of the error. Error code values less than 100 are controller fatal faults that effect the basic operation of the controller. These errors are indicated as **Err n** on the digital display and DO3 and DO4 are simultaneously activated.

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Non fatal faults appear as a flashing **E** on the digital display. Refer to Appendix A for a complete listing of error codes. To view actual error codes, the user must enter the configuration mode.

5.14.2 Power Failure

After a power failure the controller restarts using the functions configured at the time of failure and the output signal set with FCODE 20. All parameters and selected functions retain their values stored in non-volatile memory.

Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Start Value	Output signal AO1	20	0-100	%	1%	0	

If the controller was in the configuration mode, autotuning, or showing the setpoint value when the failure occurred, it will not automatically resume after the failure. Return to these conditions must be accomplished manually.

5.15 Authorization Check

Use FCODE 139 to enable an authorization check to prevent unauthorized adjustments to the controller setting.

Function	FCODE	Option	No	Default Value	Selected Value
Authori-zation	139	No authorization code set Authorization code set	1 2	1	

When the authorization check is configured, the parameters and functions may be read, but not altered. The user must perform the following steps to change functions.

- Press **PROG** button to enter the Programming mode.
- **FC 1** should appear on the display if there are no error codes.
- If error codes are present, use the **PROG** or **SET** buttons to view and remove error codes.
- To allow access for changes, press the **R** button immediately after viewing error codes and when the display reads **FC 1**. This button must be pressed to change configured functions and tuning parameters.

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- **PROG** appears on the display, the user now has full access to function and parameter changes.
- See Tuning Parameters/Configuration Changes for instructions to make appropriate changes.

Note

If the programming mode is exited before changing the authorization, you must start the procedure from the beginning.

5.16 Resetting the Controller

Use FCODE 144 and 145 to reset the controller to the default values.

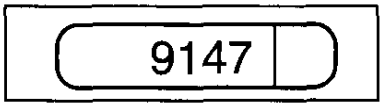
Function	FCODE	Option	No	Default Value	Selected Value
Resetting	144	Reset disabled Reset Autotuner Reset Gain Scheduling	0 1 2	0	

Caution

Pressing the STORE button when this FCODE is displayed will reset the entire controller.

Function	FCODE	Option	No	Default Value	Selected Value
Resetting	145	Reset disabled Reset configuration	0 1	0	

5.17 Software Version Number



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When addressing specific questions about the application and operation of the DPR900 Integral Controller, it is necessary to have the software version number available. Select FCODE 45 and press the **TUNE** button to view the installed software version number. The display may read 9147 for example, meaning year 1991, week 47.

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Function	Parameters	FCODE	Range	Unit	Increment	Default Value	Selected Value
Software Version (YYWW)	YY = year WW = week	45				9147	

5.18 Configuration Planning Worksheets

Worksheet 5-1 is an expanded version of the parameters section of the configuration reference card that is stored behind the hinged cover on the front panel. The expanded reference shows.

- the function code (FCODE) for selecting each parameter
- the function and parameter description
- the allowable parameter range
- the parameter units
- the increment that the increase/decrease buttons change the parameter value during manual tuning
- the default parameter value for each controller function

The horizontal lines in the worksheet separate the FCODE's by function. Within functional groupings, blank lines indicate subgroups. For example, the alarm limits group is subdivided into process value alarms and deviation alarm limits.

Worksheet 5-1 also has an empty column so that the user can enter the parameter values as he plans configuration, and record parameter values changed from the defaults. The configuration reference card also has a space for writing in these parameter values.

Worksheet 5-2 shows the FCODE, descriptive and default information for controller functions. There are blank columns for maintaining a record of selected functions and values.

Worksheet 5-3 has a space for additional notes on the use of I/O signals.

Use the worksheets in this manual as masters to produce additional copies as required.

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Worksheet 5-1. Parameters

Function	Parameter	FCODE	Range	Unit	Increment	Default Value	Selected Value
Ratio(AI2) Bias (AI2)	Gain Zero level offset	1 2	0—9.99 ± Meas.	– any	0.01 1 digit	1 0	
Process Value Alarm Limits	Limit 1 Limit 2	3 4	Meas. Meas.	any any	1 digit 1 digit	Min Max	
Deviation Alarm Limits	Negative limit Positive limit	5 6	0—(–Meas) 0—(+Meas)	any any	1 digit 1 digit	–Max +Max	
Output Signal Limits (AO1)	Lower limit Upper limit	7 8	0—100 0—100	% %	1% 1%	0 100	
Setpoint Limits	Low limit High limit	9 10	Meas. Meas.	any any	1 digit 1 digit	Min Max	
Feedforward	Gain	11	0— ±9.99	–	0.1	0	
Extra Internal Setpoint	Setpoint value	12	Meas.	any	1 digit	Min	
Setpoint Ramping	Ramping rate Ramping timer	13 14	Meas. 0—999	any min	1 digit 1 digit	Max 1 min	
Analog Input Filter	Time constant AI1 Time constant AI2 Time constant AI3 Time constant AI4	15 16 17 18	0—9999 0—9999 0—9999 0—9999	s s s s	1s 1s 1s 1s	0 0 0 0	
Forced Control Start Value Controller Fault	Output signal AO1 Output signal AO1 Output signal AO1	19 20 21	0—100 0—100 0—100	% % %	1% 1% 1%	0 0 0	
Process Value Monitoring	Max deviation Delay time	22 23	Meas. 0—999	any s	1 digit 1s	Max 10	
Measuring Range	Minimum value Maximum value Decimal point	24 25 26	± 9999 ± 9999 1 — 4	any any –	1 digit 1 digit 1	0 100 1	
Option For Output Positioning Motor	Raise/Lower Deadband Running time for 0—100% change	27 28	0—9.9 1—999	% s	0.1% 1s	0.5 60s	
Potentiometer Feedback	Current output	29	0—20	mA	0.1mA	5mA	
Actuator Feed- back Signal	Gain Zero offset	30 31	0—9.99 +100	– %	0.01 1%	1 0	
Pulse-Width Output Option	Cycle time	32	1—999	s	1s	100s	
DDC Running Time	Time for 0—100% change	33	1—999	s	1s	100s	
Gain Scheduling Breakpoints	Lower breakpoint Upper breakpoint	34 35	0—100 0—100	% %	1% 1%	33% 67%	

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Worksheet 5-1. Parameters (Continued)

Function	Parameter	FCODE	Range	Unit	Increment	Default Value	Selected Value
PID Parameter 1	Gain	36	.01—99.99	–	0.01	0.01	
	Ti (Reset)	37	OFF—9999	s	1s	1000s	
	Td (Rate)	38	0—9999	s	1s	0	
PID Parameter 2	Gain	39	.01—99.99	–	0.01	0.01	
	Ti (Reset)	40	OFF—9999	s	1s	1000s	
	Td (Rate)	41	0—9999	s	1s	0	
PID Parameter 3	Gain	42	.01—99.99	–	0.01	0.01	
	Ti (Reset)	43	OFF—9999	s	1s	1000s	
	Td (Rate)	44	0—9999	s	1s	0	
Software Version (YYWW)	YY = year WW = week	45				9147	
COMLI ID	Controller's highway address	46	1—255			1	

Worksheet 5-2. Functions

Function	FCODE	Option	No	Default Value	Selected Value
Process Type	101	Reverse (negative gain)	1	1	
		Direct (positive gain)	2		
Controller Output	102	Analog output	1	1	
		Positioning motor output (DO5)(DO6)	2		
		Pulse-width modulated output (DO5)	3		
Process Value Alarm Selection	103	Disabled	0	0	
		High/Low alarm	1		
		High alarm	2		
		Low alarm	3		
		High/High-High alarm	4		
		Low/Low-Low alarm	5		
Deviation Alarm Selection	104	Disabled	0	0	
		Enabled	1		
Output Signal Limits AO1	105	Disabled	0	0	
		Signal limits apply in Auto/Manual	1		
		Signal limits apply in Auto only	2		
Setpoint Limits	106	Disabled	0	0	
		Enabled	1		
Calculation of Process Variable	107	Disabled	0	0	
		Square root (AI1)	1		
		R = (AI1 x Ratio) + Bias	2		
		R = AI1 + (AI3 x Ratio) + Bias	3		
		R = (AI1 x Ratio) + AI3 + Bias	4		
		R = (AI1 x AI3)(Ratio) + Bias	5		

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Worksheet 5-2. Functions (Continued)

Function	FCODE	Option	No	Default Value	Selected Value
Calculation of Remote Setpoint	108	Disabled R = (AI2 x Ratio) + Bias R = AI2 + (AI3 x Ratio) + Bias R = (AI2 x Ratio) + AI3 + Bias R = (AI2 x AI3)(Ratio) + Bias	0 1 2 3 4	0	
Setpoint Ramping	109	Disabled Enabled, output on DO3-DO4 Enabled, no indicated output	0 1 2	0	
External A/M Track Switching	110	Disabled Auto/Manual/Track switching via DI1 Auto/Manual/Track switching via DI2	0 1 2	0	
External R/L Switching	111	Disabled Remote/Local switching via DI1 Remote/Local switching via DI2	0 1 2	0	
Forced Output Control	112	Disabled DI1, applies in Auto/Manual DI2, applies in Auto/Manual DI1, applies in Auto only DI2, applies in Auto only	0 1 2 3 4	0	
Controller Fault	113	Freeze output at current level Set output level to preset value	1 2	1	
Output Selector AO2	114	Disabled Process value Active setpoint Local setpoint Calculated Remote setpoint Internal output signal	0 1 2 3 4 5	0	
Output Signal Processing	115	Disabled Feedforward enabled, AI3 Feedforward enabled, AI4 Tracking, Output DI (AI4) Tracking, Signal Selector (AI4) Min selection enabled Max selection enabled	0 1 2 3 4 5 6	0	
PV Monitoring, PV/RSP Value Selection	116	Disabled PV Monitoring DO3, DO4 PV Monitoring, not using DO Min selection AI1/AI3 PV Max selection AI1/AI3 PV Min selection AI2/AI3 RSP Max selection AI2/AI3 RSP	0 1 2 3 4 5 6	0	
Data Communication	117	Disabled Not used Output signal control Remote setpoint control COMLI RS-485 protocol	0 1 2 3 4	0	

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Worksheet 5-2. Functions (Continued)

Function	FCODE	Option	No	Default Value	Selected Value
Temperature Input (AI1) (Used only when DPR100 module is used)	118	Disabled 100 ohm Platinum RTD, °C Type J Thermocouple, °C Type K Thermocouple, °C Type R Thermocouple,° Type S Thermocouple, °C 100 ohm Platinum RTD, °F Type J Thermocouple, °F Type K Thermocouple, °F Type R Thermocouple, °F Type S Thermocouple, °F	0 1 2 3 4 5 6 7 8 9 10	0	
AI1 Signal Range	119	4—20mA, 1—5V 0—20mA, 0—5V	1 2	1	
AI2 Signal Range	120	4—20mA, 1—5V 0—20mA, 0—5V 4—20mA, 1—5V, square root 0—20mA, 0—5V, square root	1 2 3 4	1	
AI3 Signal Range	121	4—20mA, 1—5V 0—20mA, 0—5V 4—20mA, 1—5V, square root 0—20mA, 0—5V, square root	1 2 3 4	1	
AI4 Signal Range	122	4—20mA, 1—5V 0—20mA, 0—5V 4—20mA, 1—5V, square root 0—20mA, 0—5V, square root	1 2 3 4	1	
Signal Range AO1	123	4—20mA output signal 0—20mA output signal	1 2	1	
Signal Range AO2	124	4—20mA output signal 0—20mA output signal	1 2	2	
DI1 Active State	125	Voltage present at input No voltage present at input	1 2	1	
DI2 Active State	126	Voltage present at input No voltage present at input	1 2	1	
DI3 Active State	127	Voltage present at input No voltage present at input	1 2	1	
DI4 Active State	128	Voltage present at input No voltage present at input	1 2	1	
DO1 (A/M) Active State	129	Open Closed	1 2	1	
DO2 (R/L) Active State	130	Open Closed	1 2	1	
DO3 (Alarm) Active State	131	Open Closed	1 2	1	
DO4 (Alarm) Active State	132	Open Closed	1 2	1	

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Worksheet 5-2. Functions (Continued)

Function	FCODE	Option	No	Default Value	Selected Value
Actuator Feedback Potentiometer	133	Not connected Connected to AI4 Connected to AI3	1 2 3	1	
Display Max Output Signal	134	100 percent 0 percent	1 2	1	
Autotuner	135	Disabled Enabled Enabled, break on alarm	0 1 2	0	
Max Output Signal Change	136	10 percent 5 percent 3 percent	1 2 3	1	
Autotune Dynamics	137	Medium, critical damping Fast, slightly underdamped Slow, slightly overdamped Deadtime	1 2 3 4	1	
Gain Scheduling Reference (GSref)	138	Disabled Output signal Process value Signal is AI3 Signal is AI4	0 1 2 3 4	0	
Authorization	139	No authorization code set Authorization code set	1 2	1	
Setpoint Tracking (LSP)	140	Disabled Enabled, tracks remote setpoint RSP/ESP Enabled, tracks PV in Manual	1 2 3	1	
Setpoint with Computer Control	141	Extra internal setpoint External remote setpoint	1 2	1	
Automatic Setpoint Switching	142	Disabled Switches to remote setpoint Switches to extra internal setpoint	1 2 3	1	
Normal Display	143	Display process value Display AI1, in percent Display AI2, in percent Display AI3, in percent Display AI4, in percent	1 2 3 4 5	1	
Resetting	144	Reset disabled Reset Autotuner Reset Gain Scheduling	0 1 2	0	
Resetting	145	Reset disabled Reset configuration	0 1	0	

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Worksheet 5-3. Signal Allocations

Signal	Used for	Notes
AI1	Process Value:	
AI2	Remote Setpoint:	
AI3		
AI4		
AO1	Output Signal, Potentiometer Feedback:	
AO2	Output of:	
DI1	Switching to:	
DI2	Switching to:	
DI3	Control of:	
DI4	Control of:	
DO1	Indicate Auto/Manual:	
DO2	Indicate Local/Remote:	
DO3	Indicate:	
DO4	Indicate:	
DO5	Pulse Output:	
DO6	Pulse Output:	

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Appendix A

A.1 Fault Codes

The DPR900 Integral Controller has two fault levels:

- Warnings
- Fatal errors

A.1.1 Warnings

Faults are shown by a flashing **E** in the digital display, and digital outputs DO3 and DO4 are simultaneously activated. These are the same outputs used for normal alarm indications.

Display the fault code, using the following procedure:

- Press the PROG button.
- The fault code will appear on the digital display.
- Press the yellow increase button.
- If more than one fault code has been generated, the others are shown as the increase button is pressed.
- Exit the fault code indication mode by pressing PROG or TUNE button. PROG selects the programming mode and leaves the fault codes stored. TUNE selects the Programming mode and erases all current fault codes.
- Press the PROG button to exit the Programming mode.

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Fault Code	Meaning	Precautions
E 101	Fault detected while monitoring process value with sensor 1. Sensor 2 has taken over control.	Ensure that sensor 1 is connected and operating properly.
E 102	Fault detected while monitoring process value with sensor 2. Sensor 1 has taken over control.	Ensure that sensor 2 is connected and operating properly.
E 103	Output signal failure at analog output 2 (AO2).	Check current path between pins 11 and 12. Check connections and receiving device. Check that load is less than 650 ohms. Check dc supply for overload (max 100mA). Check supply voltage is not less than 10% of nominal. Check FCODE 124 set to Option 2 if AO2 is not in use.
E 104	Controller has not been tested. It was either delivered untested or the parameter and function EEPROM has been exchanged or corrupted since delivery.	Contact your Fisher representative.
E 105	Difference between setpoint value and process value when autotuning started is too large. Tuning started with output signal too close to 0% or 100%. Tuning function is aborted.	Bring process value and setpoint value closer to each other, then restart autotuner function. Retune at level between 10 and 90%.
E 106	**WARNING:** Process noise level too high or process not stable enough for autotuner function to operate properly. This is only a warning, the controller continues to tune, but results may not be optimum.	A process load upset during early tuning. Switch controller to manual mode and stabilize process, or filter the process value. Retune.
E 107	**WARNING:** Disturbance has occurred in the process during a tuning procedure. Tuning function continues, but results may not be optimum.	Check PID parameters and possibly retune the controller as the parameters may not be optimum.
E 108	Tuning procedure has exceeded the 32-hour maximum limit. This is a fatal tuning error and tuning has been aborted.	None required. Controller automatically returns to the state it was in before the user started the autotuner.

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Fault Code	Meaning	Precautions
E 109	Tuning has been stopped due to an alarm condition being present (configured by FCODE 135).	None required. Controller automatically returns to the state it was in before the user started the autotuner.
E 110	Error in analog correction factor as calibrated at the factory is detected. The controller still functions, but factory-set default values are now being used in place of the factory-calibrated values.	To clear, this unit must be returned to the factory for recalibration.

A.1.2 Fatal Errors



Note

A fatal error affects the function of the control loop and must be acted upon immediately.

Fatal errors are shown as **Err X** in the digital display. When possible, fatal errors must be corrected before the controller will function normally.

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Fault Code	Meaning	Precautions
Err 1	Checksum fault in the internal program memory. DO3 and DO4 are activated. The unit sends out a user-defined output signal (FCODE 23) or freezes the output signal (FCODE 113). Unit switches to Manual. It is possible to control the output signal with the increase/decrease buttons, but the Manual mode cannot be exited. This error can occur if the installation was done improperly.	Reset the configuration using FCODE 145, then reconfigure the controller. Controller should show 0.0 on the digital display. Check the installation. If Err 1 recurs, contact your Fisher representative. Record the configured functions and tuning parameters from the digital display in the Programming mode.
Err 2	Checksum fault in the controller program. DO3 and DO4 are activated. The unit sends out a user-defined output signal (FCODE 23) or freezes the output signal (FCODE 113). Behavior of the controller cannot be guaranteed. It is possible to control the output signal with the increase/decrease buttons, but the Manual mode cannot be exited.	Contact your Fisher representative.
Err 3	Thermocouple failure when using the DPR100 temperature module. Possible thermocouple burnout. DO3 and DO4 are activated. The unit sends out a user-defined output signal (FCODE 23 or freezes the output signal (FCODE 113) and switches to Manual. It is possible to control the output signal with the increase/decrease buttons, but the Manual mode cannot be exited. If the failure is cleared, the controller returns to normal operation.	Ensure that the sensor is working and check the signal connections between the sensor and the DPR100 module.
Err 4	Output signal failure at AO1. The output has dropped out and the unit switches to Manual. DO3 and DO4 are activated. The output signal cannot be controlled by the increase/decrease buttons. If the failure is cleared, the controller returns to normal operation.	Ensure that connections to the AO terminals are secure, and check the receiving device for proper operation. Check for current path between pins 9 and 10 on terminal block. Check that load is less than 650 ohms. Check that dc supply is not overloaded, max 100mA. Check that supply voltage is not less than 10% of nominal.
Err 5	Communications failure with the display panel circuits. DO3 an DO4 are activated. The unit sends out a user-defined output signal (FCODE 23) or freezes the output signal (FCODE 113). If possible, digital display shows **Err 5** or goes blank.	Internal hardware error. Contact your Fisher representative.

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A.2 Main Power Failure

When power is restored, the DPR900 will continue with the function it had before the failure, using the output signal determined by the setting of FCODE 20. With 3-state pulse output, the actuator feedback must be present for this to operate.

If the controller was in the Programming mode, performing Autotuning or displaying setpoint in the digital display, these functions will not be resumed automatically after a power failure. If required, they must be selected again.

All parameters and functions which were selected during configuration will remain stored in memory.

On restart, the software version number will be displayed for about 3 seconds in the digital display.

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Appendix C

C.1 Field Replaceable Parts

The following is a list of field replaceable parts that the user may want on hand as spares. These parts can be ordered from your Fisher Controls representative.

Part Description	Part Number
Front Display Panel (w/gasket and display border	DPR900SPR-DISP
Engineering Units Labels	DPR900SPR-Label
Spare Mounting Cassette (All Voltages)	DPR900SPR-120V
Adjustable Mounting Bracket (2 required for thick panel mounting)	11B9245
Spare Parts Package: 1 Hinge flap 2 Hinge pins 1 Hinge spring 2 Retainer springs 1 Cable tie plate 3 Protection plates 1 Wiring label	12B9958

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	Yes	No
5. Is the information organized in a logical manner?	☐	☐
6. Can you find specific information in a reasonable period of time?	☐	☐
7. Does the manual describe the operation of the product.	☐	☐
8. Is information adequately cross-referenced?	☐	☐
9. Is the information easy to understand?	☐	☐
10. Is there too much information?	☐	☐
11. Are there sufficient examples and illustrations?	☐	☐

	Excellent	Average	Poor
12. How do you rate the *overall usability* of this manual?	☐	☐	☐

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